Modern Lectures on QFT

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Abstract

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1 Scale-Invariant Physics

These notes summarize the Wilsonian renormalization-group viewpoint and its application to the Ising universality class. The prose has been lightly reconstructed; the equations have been normalized and checked for consistency.

1.1 Wilsonian RG in QFT

Consider a Euclidean scalar theory with cutoff Λ:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{\lambda}{4!}\varphi^4, \tag{1}$$

$$Z = \int [\mathcal{D}\varphi]_\Lambda \exp\Big(-\int d^d x \mathcal{L}[\varphi]\Big), \quad [\mathcal{D}\varphi]_\Lambda = \prod_{|k|<\Lambda} d\varphi(k), \tag{2}$$

with Fourier convention

$$\varphi(x) = \int \frac{d^d k}{(2\pi)^d} e^{ik\cdot x} \varphi(k). \tag{3}$$

One may worry about changing into the fourier space may cause a change of measure, indeed, yet long as we focus on physical observables, the change of measure does not affect the final results.

> **Remark**
>
> Here in Wilsonian's picture, all QFT are defined up to a cut off in the momentum space or equivalently a lattice spacing in the position space.

1.1.1 Integrating out UV Modes

Now we want to consider how a theory changes as we lower the energy cut off. Now lower the cutoff from Λ to $\rho\Lambda$, where $0 < \rho < 1$, and split

$$\varphi(k) = \varphi_{\text{IR}}(k) + \hat{\varphi}(k), \tag{4}$$

where φ_{IR} has support for $|k| < \rho\Lambda$ and $\hat{\varphi}$ has support for $\rho\Lambda \leq |k| < \Lambda$. Then we assume that m, λ are small and we integrate out the high-momentum modes $\hat{\varphi}$:

$$Z = \int [\mathcal{D}\varphi]_{\rho\Lambda} \int [\mathcal{D}\hat{\varphi}] \exp\left(-\int d^d x \mathcal{L}[\varphi + \hat{\varphi}]\right) = \int [\mathcal{D}\varphi]_{\rho\Lambda} \exp\left(-\int d^d x \mathcal{L}_{\text{eff}}[\varphi]\right). \tag{5}$$

The effective Lagrangian may in principal contains every local operator allowed by the symmetries:

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}(1 + \Delta Z)(\partial\varphi)^2 + \frac{1}{2}(m^2 + \Delta m^2)\varphi^2 + \frac{1}{4!}(\lambda + \Delta\lambda)\varphi^4 + \Delta C(\partial\varphi)^4 + \Delta D\varphi^6 + ... \tag{6}$$

All induced coefficients depend on the blocking parameter ρ. Now we see that the original theory if we integrate out the high momentum modes first we generates an equivalent theory with a lower cutoff and new lagrangian. Notice that the new lagrangian is not a completely new theory but rather a effective theory of the original theory if we only consider the low energy physics.

1.1.2 Flow on Coupling Space

Now we want to interpret the new effective theory as a theory with a change of the coupling from the original theory (while preserving the cut off and the Kinetic term). To do this we need to restore the original cutoff by defining

$$x' = \rho x, \quad k' = \frac{k}{\rho}, \quad \partial_\mu = \rho\partial'_\mu. \tag{7}$$

We also want the Kinetic term to be canonically normalized thus we can fix a normalization of the field and only have the coupling to flow.

$$\varphi' = \sqrt{\rho^{2-d}(1 + \Delta Z)}\varphi. \tag{8}$$

The action therefore becomes

$$\int d^d x' \left[\frac{1}{2}(\partial'\varphi')^2 + \frac{1}{2}m'^2\varphi'^2 + \frac{\lambda'}{4!}\varphi'^4 + C'(\partial'\varphi')^4 + ...\right], \tag{9}$$

where

$$\begin{aligned} m'^2 &= (m^2 + \Delta m^2)(1 + \Delta Z)^{-1}\rho^{-2}, \\ \lambda' &= (\lambda + \Delta\lambda)(1 + \Delta Z)^{-2}\rho^{d-4}, \\ C' &= (C + \Delta C)(1 + \Delta Z)^{-2}\rho^d. \end{aligned} \tag{10}$$

> **Remark**
>
> Notice that until now we have not made any approximation. But obviously the coefficients for the new effective lagrangian are quite impossible to compute.

However, now we focus on leading order in m^2, λ, C and we have:

$$m'^2 = \rho^{-2} m^2, \qquad \lambda' = \rho^{d-4}\lambda, \qquad C' = \rho^d C. \tag{11}$$

We may see that different couplings behave differently under the RG flow. The mass term m^2 grows as we lower the cutoff, the λ term grows for $d < 4$ and shrinks for $d > 4$, and the C term shrinks for all $d > 0$. Thus we call:

- Relevant operators: m^2 and λ for $d < 4$.
- Irrelevant operators: C and λ for $d > 4$.
- Marginal operators: λ for $d = 4$. In fact this need detailed analysis, in dim reg we in fact work in $d = 4 - \varepsilon$ and we find that λ is marginally relevant for $\varepsilon > 0$.

1.1.3 RG Flow and Fixed Points

The flow of the couplings can be described by a function called the beta function:

$$\mu \frac{dg}{d\mu} = \beta(g). \tag{12}$$

with μ the energy scale and g the coupling. We may find out that there are some fixed point of the RG flow, which have some interesting behaviors.

YL: [Diagram in hand written note!]

1.2 RG in StatMech

For RG in StatMech, it is much more calculable compared to QFT (where in fact we seldome directly calculate the RG flow of Wilsonian coupling.) Now we take Ising model as a simplest model for an illustration. For lattice with spins $s_i \in \{+1, -1\}$ on each site:

$$H = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i, \qquad P[\{s\}] = \frac{1}{Z} e^{-\beta H[\{s\}]}, \qquad Z = \sum_{[\{s\}]} e^{-\beta H[\{s\}]}, \tag{13}$$

where $\beta = \frac{1}{T}$. At $h = 0$, the low-temperature phase has nonzero spontaneous magnetization, while the high-temperature phase is disordered. Near the critical temperature,

$$M \sim (T_c - T)^\beta \tag{14}$$

on the ordered side. Here the exponent β should not be confused with inverse temperature. (This is experimentally true and we will see how to understand this.)

1.2.1 Block Spin Transformation

In RG of QFT we integrate out the high momentum modes and see how the theory changes. Now in SM, it is much convenient to stay in position space and we can make the lattice spacing larger and see how the theory changes. To do this we integrate out the blocks of spins to make it a coarse-grained lattice. This is realized by **Block Spin Transformation**:

$$T(s'; s_1, ..., s_q) = \begin{cases} 1 \text{ if } s' \sum_{i=1}^q s_i > 0 \\ 0 \text{ otherwise} \end{cases}. \tag{15}$$

This means that if most of the spins in a block are up, then the coarse-grained spin is up, and vice versa or we throw away the contribution.

YL: [Diagram for illustration!]

The coarse-grained Hamiltonian is defined by

$$e^{-\beta H'[\{s'\}]} = \sum_{[\{s\}]} \prod_{\text{blocks}} T\big(s'; s_1, ..., s_q\big) e^{-\beta H[\{s\}]}. \tag{16}$$

and the new partition function is

$$\begin{aligned} Z' &= \sum_{[\{s'\}]} e^{-\beta H'[\{s'\}]} = \sum_{[\{s'\}]} \sum_{[\{s\}]} \prod_{\text{blocks}} T\big(s'; s_1, ..., s_q\big) e^{-\beta H[\{s\}]} \\ &= \sum_{[\{s\}]} e^{-\beta H[\{s\}]} \sum_{[\{s'\}]} \prod_{\text{blocks}} T\big(s'; s_1, ..., s_q\big). \end{aligned} \tag{17}$$

Since $\sum_{s'} T = 1$, one has $Z' = Z$. We can see that the partition function is never changed, we cut integrate out the blocks first and the integrate out the coarse-grained spins later, which is consist with integrating out the highier momentum modes in QFTs.

One may expect a second step of rescaling to restore the original lattice spacing and the Kinetic term to have the field normalized. However, the Ising CFT is independent of the lattice spacing and the field is already normalized in the definition of the BST to have σ fixed to be $+1$ or -1. Thus we do not need to do any rescaling.

From the new Hamiltonian $H'[\{s'\}]$, reading off the new couplings gives an RG map

$$(J', h', ...) = \mathrm{RG}[(J, h, ...)]. \tag{18}$$

1.2.2 One-dimensional decimation

For 1D Ising model, we can do the RG transformation exactly. For $d = 1$ and $h = 0$, choose every two spins to live in a block and integrate out the block with a special BST which only sees the even number blocks and simply ignore the first one. The BST is:

$$T(s'; s_1, s_2) = \delta_{s', s_2}, \qquad H = -J \sum_i s_i s_{i+1} - c \sum_i 1. \tag{19}$$

> **Remark**
>
> Here in the original theory we add a term of $-c$ to the Hamiltonian, which is to nromalize out the vaccum energy.

Summing over inside the blocks gives the coarse-grained Hamiltonian as

$$\begin{aligned} e^{-\beta H'[\{s_{\text{even}}\}]} &= \sum_{s_{\text{odd}}} e^{\beta\left[J \sum_{\text{even}} (s_i s_{i+1} + s_{i+1} s_{i+2}) + 2 \sum_{\text{even}} c\right]} \\ &= \prod_{\text{even}} \left[e^{\beta(J(s_i + s_{i+2}) + 2c)} + e^{\beta(-J(s_i + s_{i+2}) + 2c)} \right]. \end{aligned} \tag{20}$$

Notice that here s_i, s_{i+2} are both fixed number. In the final form, the first term is given by $s_{i+1} = 1$ and the second term is given by $s_{i+1} = -1$. Matching this to $e^{J' s_i s_{i+2} + c'}$, and redefining $K = \beta J$, $K' = \beta J'$, and $C = \beta c$, we have:

$$K' = \frac{1}{2} \ln(\cosh(2K)), \qquad C' = 2C + \frac{1}{2} \ln(4 \cosh(2K)). \tag{21}$$

With $x = \tanh K$, $0 \le x < 1$,

$$x' = \mathrm{RG}(x) = x^2. \tag{22}$$

Thus the unstable zero-temperature fixed point $x = 1$ flows toward the stable infinite-temperature fixed point $x = 0$.

1.3 Linearized flow near a fixed point

Let K_a be a set of couplings and $K'_a = \mathrm{RG}(K_a)$ be the RG transformation. We define a fixed point K^*_a as a point that is invariant under the RG transformation:

$$K^*_a = \mathrm{RG}(K^*_a). \tag{23}$$

Let K^*_a be a fixed point. Linearizing the RG transformation gives

$$K'_a - K^*_a = \left[\frac{\partial K'_a}{\partial K_b}\right]_{K=K^*} (K_b - K^*_b) \equiv T_{ab}(K_b - K^*_b). \tag{24}$$

Let v^i_a be left eigenvectors (i here is the label of eigenvectors),

$$v^i_a T_{ab} = \lambda_i v^i_b, \tag{25}$$

and define scaling variables

$$u_i = v^i_a (K_a - K^*_a). \tag{26}$$

We now can see that the RG transformation is diagonalized in the u_i basis: Under one RG transformation, the new scaling variable is

$$\begin{aligned} u'_i &= v^i_a (K'_a - K^*_a) \\ &= v^i_a T_{ab}(K_b - K^*_b) + \mathcal{O}\Big((K - K^*)^2\Big) \\ &= \lambda_i v^i_b (K_b - K^*_b) + \mathcal{O}\Big((K - K^*)^2\Big) \\ &= \lambda_i u_i + \mathcal{O}(u^2). \end{aligned} \tag{27}$$

Therefore, to linear order around the fixed point, the RG transformation is diagonal in the scaling-variable basis:

$$u'_i = \lambda_i u_i. \tag{28}$$

After one blocking step, we commonly use b to denote the linear size of the block and use it to characterize how the variable transform as we change the scale.

$$u'_i = \lambda_i u_i \equiv b^{y_i} u_i, \qquad b^d = \text{number of spins per block.} \tag{29}$$

where d is the spatial dimension. Then we can now see how different parameters behave under the RG transformation.

- If $y_i > 0$, then u_i grows under RG and the corresponding operator is relevant.
- If $y_i < 0$, then u_i shrinks under RG and the corresponding operator is irrelevant.
- If $y_i = 0$, then u_i is unchanged under RG and the corresponding operator is marginal. If the deformation of

This is in fact related to the QFT picture, for a operator that perturb the fixed-point action is $\sum_i c_i \mathcal{O}_i$, the scaling dimension of $\mathcal{O}_i$ is:

$$\Delta_i = d - y_i. \tag{30}$$

1.4 Scaling Hypothesis

Most observable in StatMech system can be given from the free energy. Thus the behavior of the free energy under RG transformation near a fixed point is of great interest, it may give us behavior of observables near the critical points in terms of the couplings. The free energy density is defined as:

$$f = -\frac{1}{\beta N} \log Z, \tag{31}$$

Under RG transformation, the partition function is invariant, thus the free energy density transforms as:

$$f' = -\frac{1}{\beta N'} \log Z = b^d f \tag{32}$$

remember that b^d is the number of spins per block. The free energy density can be split into:

$$f(\{K, c\}) = c + \hat{f}(\{K\}), \tag{33}$$

We split out the vacuum energy density c since it is irrelevant to most observables. In general we have:

$$c' = b^d(c + g(\{K\})) \tag{34}$$

Thus under RG transformation, we have the following relation:

$$\hat{f}(\{K\}) = g(\{K\}) + b^{-d}\hat{f}(\{K'\}) \tag{35}$$

We now split the free energy density into singular and regular parts. We mostly care about the singular part since it is the one that gives us the signal of the second order phase transition. Normally, the function $g(\{K\})$ is regular and thus under RG the singular part of the free energy density transforms as:

$$f_s(K) = b^{-d} f_s(K'). \tag{36}$$

Now we focus on the case that we have a fixed point K^* and we are **near the fixed point**. We assume now we only have thermal and magnetic scaling variables u_t and u_h, (Just assume other irrelevant variables goes to 0 near the fixed point), we have:

$$f_s(u_t, u_h) = b^{-d} f_s(b^{y_t} u_t, b^{y_h} u_h). \tag{37}$$

After n iterations,

$$f_s(u_t, u_h) = b^{-nd} f_s(b^{ny_t} u_t, b^{ny_h} u_h). \tag{38}$$

Choose to end on some n so that $|b^{ny_t} u_t| = u_{t0}$. Then plugging in n in terms of other quantities gives:

$$f_s(u_t, u_h) = \left|\frac{u_t}{u_{t0}}\right|^{\frac{d}{y_t}} f_s\left(\mathrm{sgn}(u_t) u_{t0}, u_h \left|\frac{u_t}{u_{t0}}\right|^{-\frac{y_h}{y_t}}\right). \tag{39}$$

Using reduced temperature $t = \frac{T - T_c}{T_c}$ and the relation $u_t = \frac{t}{t_0}, u_h = \frac{h}{h_0}$ (we here won't justify why this is true, one can assume as a fact for Ising), this is commonly written as

$$f_s(t, h) = \left|\frac{t}{t_0}\right|^{\frac{d}{y_t}} \Phi\left(\left(\frac{h}{h_0}\right)\left|\frac{t}{t_0}\right|^{-\frac{y_h}{y_t}}\right). \tag{40}$$

Consequently, taking derivatives gives the behavior of other observables:

$$C \sim \partial_t^2 f_s \sim |t|^{-\alpha}, \quad M \sim \frac{\partial f_s}{\partial h} \mid_{h=0} \sim (-t)^{\beta}, \quad \chi \sim \frac{\partial M}{\partial h} \mid_{h=0} \sim |t|^{-\gamma}, \tag{41}$$

with

$$\alpha = 2 - \frac{d}{y_t}, \qquad \beta = \frac{d - y_h}{y_t}, \qquad \gamma = \frac{2y_h - d}{y_t}. \tag{42}$$

We can see that the critical exponents are determined by y_t and y_h, which are determined by the scaling properties of the coupling near the fixed point. And they are related to each other, cause they are not independent.

In a fixed spatial dimension, the Ising lattice model, uniaxial magnets, the liquid–gas critical point, and one-component scalar φ^4 theory can belong to the same Ising universality class, that flow to the same fixed point at IR and thus have the same critical behavior.

2 Conformal Transformations

We now focus on a theory on the RG fixed point. One can deduce that the RG fixed point definition of the correlation function will lead to the following identity:

$$\langle \mathcal{O}_1(\lambda x_1)...\mathcal{O}_n(\lambda x_n)\rangle = \lambda^{-\Delta_1-...-\Delta_n}\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle. \tag{43}$$

One may find that this is exactly the definition of symmetry under scaling transformations of certain covariant fields. In fact, at RG fixed points we often have a larger symmetry group, the conformal group. In this section we will discuss the conformal group and its implications for correlation functions.

2.1 Conformal transformations

2.1.1 The conformal Killing equation

A conformal transformation is a coordinate transformation $x \to x'$ for which

$$g'_{\mu\nu}(x') = \Omega(x) g_{\mu\nu}(x). \tag{44}$$

> **Remark**
>
> This is a very generalized equation, without having the metric fixed. Starting now, we will fix a metric and use the flat Euclidean metric $\eta_{\mu\nu} = (1, 1, 1, ...)$, which is the identity matrix. Notice that the Minkowski metric will also lead to a similar result.

For an infinitesimal map

$$x'^{\mu} = x^{\mu} + \varepsilon^{\mu}(x), \tag{45}$$

the Jacobian and its inverse are

$$\frac{\partial x'^{\mu}}{\partial x^{\alpha}} = \delta^{\mu}_{\alpha} + \partial_{\alpha}\varepsilon^{\mu}, \qquad \frac{\partial x^{\mu}}{\partial x'^{\alpha}} = \delta^{\mu}_{\alpha} - \partial_{\alpha}\varepsilon^{\mu}. \tag{46}$$

To first order, we have:

$$g'_{\mu\nu} = g_{\mu\nu} - \partial_{\mu}\varepsilon_{\nu} - \partial_{\nu}\varepsilon_{\mu}. \tag{47}$$

Writing $\Omega = 1 - f$, one obtains the conformal Killing equation

$$\partial_{\mu}\varepsilon_{\nu} + \partial_{\nu}\varepsilon_{\mu} = f(x) g_{\mu\nu}, \qquad f(x) = \frac{2}{d}\partial_{\mu}\varepsilon^{\mu}. \tag{48}$$

In flat space, differentiating and permuting indices gives

$$2\partial_{\mu}\partial_{\nu}\varepsilon_{\rho} = \eta_{\rho\nu}\partial_{\mu}f + \eta_{\rho\mu}\partial_{\nu}f - \eta_{\mu\nu}\partial_{\rho}f. \tag{49}$$

Tracing and differentiating imply

$$2\partial^2\varepsilon_{\rho} = (2-d)\partial_{\rho}f, \tag{50}$$

$$(2-d)\partial_{\mu}\partial_{\nu}f = \eta_{\mu\nu}\partial^2 f. \tag{51}$$

Hence

$$\begin{cases} \text{no local constraint on } f & d = 1 \\ \partial^2 f = 0 & d = 2 \\ \partial_{\mu}\partial_{\nu}f = 0 & d > 2. \end{cases} \tag{52}$$

2.1.2 Finite conformal transformations

For $d > 2$, $f = A + B_\mu x^\mu$. We then take the ansatz of ε^μ given as:

$$\varepsilon^\mu = a_\mu + b_{\mu\nu}x^\nu + c_{\mu\nu\rho}x^\nu x^\rho \quad (2.11) \tag{53}$$

Here $c_{\mu\nu\rho} = c_{\mu\rho\nu}$. For now a_μ, $b_{\mu\nu}$, and $c_{\mu\nu\rho}$ are arbitrary constants. Finally, we can plug this form of ε^μ into the conformal killing equation and require the condition to be satisfied for all x. This leads to the following:

- a_μ is free of constraints, and corresponds to infinitesimal translations.
- $b_{\mu\nu} = \alpha\eta_{\mu\nu} + m_{\mu\nu}$, with $m_{\mu\nu} = -m_{\nu\mu}$. The trace part corresponds to an infinitesimal scale transformation. $m_{\mu\nu}$ corresponds to an infinitesimal Lorentz transformation/rotation.
- $c_{\mu\nu\rho} = \eta_{\mu\rho}b_\nu + \eta_{\mu\nu}b_\rho - \eta_{\nu\rho}b_\mu$, for a constant vector b_μ.

The corresponding finite transformations are

$$\begin{aligned} \text{translation:} \quad & x'^\mu = x^\mu + a^\mu, \\ \text{rotation/Lorentz:} \quad & x'^\mu = M^\mu_\nu x^\nu, \\ \text{dilatation:} \quad & x'^\mu = \lambda x^\mu, \\ \text{SCT:} \quad & x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b\cdot x + b^2x^2}. \end{aligned} \tag{54}$$

An important transformation is called inversion is

$$I: \quad x'^\mu = \frac{x^\mu}{x^2}, \tag{55}$$

and an SCT can be written as inversion–translation of $-b$ –inversion.

2.2 Representation

2.2.1 Representation on Functions

Definition 2.1 **Scalar function**

For a scalar function with zero intrinsic weight, we define it as transforming under the following rule with respect to any spacetime transformation $x \to x'$:

$$f'(x') = f(x), \tag{56}$$

Definition 2.2 **Generator**

For a transformation acting on a function, a generator G_a is defined by following with some real parameter ω_a:

$$f'(x) = e^{-i\omega_a G_a} f(x). \tag{57}$$

For a scalar function the generator for arbitrary transformation can be easily derived: We consider an infinitesimal version of a transformation, which is given by:

$$x'^\mu = x^\mu + \omega_a A^\mu_a(x), \qquad A^\mu_a(x) = \frac{\partial x'^\mu}{\partial \omega_a}\,|_{\omega=0}, \tag{58}$$

then we have:

$$
\begin{aligned}
f'(x) - f(x) &= -i\omega_a G_a f(x) = f(x - \omega_a A_a(x)) - f(x) \\
&= -\omega_a A_a^\mu(x)\partial_\mu f(x).
\end{aligned} \tag{59}
$$

$$
iG_a f(x) = A_a^\mu(x)\partial_\mu f(x). \tag{60}
$$

Then with certain conventions, one can easily find the generators of the conformal group on a scalar function:

$$
\begin{aligned}
P_\mu &= -i\partial_\mu, \\
L_{\mu\nu} &= i\big(x_\mu\partial_\nu - x_\nu\partial_\mu\big), \\
D &= -ix^\mu\partial_\mu, \\
K_\mu &= -i\big(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\big).
\end{aligned} \tag{61}
$$

Notice that the definition of the generators are convention dependent, however, $\omega_a G_a$ is convention independent. To be more concrete, we choose the convention that ω_a is related to the parameters of the finite conformal transformation as:

$$
\omega_a G_a = a^\mu P_\mu + \frac{1}{2}\theta^{\mu\nu}L_{\mu\nu} + \lambda D + b^\mu K_\mu, \qquad \theta^{\mu\nu} = -\theta^{\nu\mu}. \tag{62}
$$

The parameters have the following geometric meanings:

- a^μ is the translation distance: $x'^\mu = x^\mu + a^\mu$.
- $\theta^{\mu\nu}$ specifies an infinitesimal rotation in Euclidean spacetime, or a rotation/boost in Minkowski spacetime. Its related to the coordinate transformation by the canonical exponential map:

$$
x'^\mu = e^{\frac{1}{2}\theta^{\rho\sigma}L_{\rho\sigma}}x^\mu \quad L_{\rho\sigma} = x_\rho\delta^\mu_\sigma - x_\sigma\delta^\mu_\rho. \tag{63}
$$

- λ is the logarithm of the dilation factor: $x'^\mu = e^\lambda x^\mu$.
- b^μ specifies a special conformal transformation. It has the dimension of inverse length. The corresponding finite transformation is

$$
x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b_\nu x^\nu + b^2x^2}. \tag{64}
$$

Together, these parameters give the infinitesimal coordinate transformation

$$
x'^\rho = x^\rho + a^\rho + \theta^{\rho\nu}x_\nu + \lambda x^\rho + 2(b_\nu x^\nu)x^\rho - b^\rho x^2 + O(\omega^2). \tag{65}
$$

For example, a finite dilatation gives

$$
f'(r') = e^{-i\alpha D}f(r') = e^{-\alpha r'\partial_{r'}}f(r') = f(e^{-\alpha}r') = f(r), \tag{66}
$$

where $r' = e^\alpha r$. by exponentialing the derivative we mean it by Taylor expansions.

2.2.2 Conformal algebra

With the conventions above, we can find an algebra of the conformal generators.

$$
\begin{aligned}
\big[D, P_\mu\big] &= iP_\mu, \quad \big[D, K_\mu\big] = -iK_\mu, \quad \big[K_\mu, P_\nu\big] = 2i\big(\eta_{\mu\nu}D - L_{\mu\nu}\big), \\
\big[L_{\mu\nu}, P_\rho\big] &= -i\big(\eta_{\mu\rho}P_\nu - \eta_{\nu\rho}P_\mu\big), \quad \big[L_{\mu\nu}, K_\rho\big] = -i\big(\eta_{\mu\rho}K_\nu - \eta_{\nu\rho}K_\mu\big), \\
\big[L_{\mu\nu}, L_{\rho\sigma}\big] &= -i\big(\eta_{\mu\rho}L_{\nu\sigma} - \eta_{\mu\sigma}L_{\nu\rho} - \eta_{\nu\rho}L_{\mu\sigma} + \eta_{\nu\sigma}L_{\mu\rho}\big). \\
\big[P_\mu, P_\nu\big] &= \big[K_\mu, K_\nu\big] = [D, D] = \big[D, L_{\mu\nu}\big] = 0,
\end{aligned} \tag{67}
$$

Defining

$$J_{\mu\nu} = L_{\mu\nu}, \quad J_{-1,\mu} = \frac{1}{2}(P_\mu - K_\mu), \quad J_{0,\mu} = \frac{1}{2}(P_\mu + K_\mu), \quad J_{-1,0} = D, \tag{68}$$

one recognizes the algebra $\mathrm{so}(d+1,1)$ in Euclidean signature. If we consider Minkwoski spacetime with signature $(-,+,+,+,...)$, the algebra is $\mathrm{so}(d,2)$.

> **Remark**
>
> Above discussion is all in Euclidean spacetime, however, one can prove that the Minkowski version is just changing the metric.

> **Remark**
>
> Why functions instead of the defining representation? This question in fact we have encountered in Poincare representation. Its hard to get the commutation of the spacetime translation and the lorentz transformation using the defining representation. Thus we often just use representation on functions or an abstract representation on Hilbert space to get the algebra.

2.2.3 Action on local operators

Now we consider the action of the conformal group on the Hilbert space. For a quantum field theory, the symmetry transformations are represented by unitary operators on the Hilbert space (and generators are Hermite operators, at least for Lorenzian Theory a theory in Euclidean spacetime may be more subtle to consist with the Wick rotation). Consider a field that transforms under some transformation as:

$$x' = F(x), \qquad \varphi'(x') = F(\varphi(x)), \tag{69}$$

> **Definition 2.3** **Quantum generator**
>
> For a quantum operator, it is assumed to be operator (maybe unitary or not) and it act on local fields as:
>
> $$\varphi'(x) = U^{-1}\varphi(x)U, \qquad U = e^{-i\omega_a \hat{G}_a}. \tag{70}$$
>
> Now $\hat{G}_a$ is the quantum generators of the theory and is assumed to be a Hermite operator.

> **Remark**
>
> In lorentzian theory, the requirement of the probability conservation will tell us that the quantum operators should be unitary. However, in Euclidean theory we don't have a notion of time, thus we don't need requirement of unitary. However, we often want the Euclidean theory to have a consist wick rotate lorentzian conterpart. Such requirement will impose conditions on the properties of the quantum operators.

Here we will define ω_a just as the parameters we used for the transformation on scalar functions.

> **Remark**
>
> This relation should be understood as a definition for a quantum generator. In QFT canonical quantization scheme we often find the quantum generators are the conserved charges exactly which lead to this convention.

The infinitesimal action WLOG can contain both an orbital part and a finite-dimensional spin part:

$$\left[\hat{G}_a, \varphi(x)\right] = -G_a\varphi(x) + iR(G_a)\varphi(x). \tag{71}$$

For the momentum generators, we have:

$$\varphi'(x) = e^{ia\cdot\hat{P}}\varphi(x)e^{-ia\cdot\hat{P}}, \quad \varphi'(x) = \varphi(x-a). \quad \varphi(x+a) = e^{-ia\cdot\hat{P}}\varphi(x)e^{ia\cdot\hat{P}}. \tag{72}$$

Thus we have:

$$\varphi(x) = e^{-ix\cdot\hat{P}}\varphi(0)e^{ix\cdot\hat{P}}, \tag{73}$$

> **Remark**
>
> This might look a bit different from the books using $(+,-,-,-)$ conventions, here we are using $(+,+,+,+)$ conventions. This note is self consistent with this convention.

The infinitesimal action of the translation generator is then:

$$\left[\hat{P}_\mu, \varphi(x)\right] = i\partial_\mu\varphi(x). \tag{74}$$

Thus, we may see that we only need to define the action of conformal generators on the local operators at the origin, their commutation relation will tell us how they act on local operators at any points.

2.2.4 Primary and Descendant Fields

Definition 2.4 **Primary Fields**

We define a certain series of local operators φ_α to commute with the generators as:

$$\left[\hat{D}, \varphi_\alpha(0)\right] = i\Delta\varphi_\alpha(0), \quad \left[\hat{L}_{\mu\nu}, \varphi_\alpha(0)\right] = i\left(S_{\mu\nu}\right)_{\alpha\beta}\varphi_\beta(0). \quad \left[K_\mu, \varphi_\alpha(0)\right] = 0. \tag{75}$$

The commutation relations beyond the origin can be derived from the algebra with the translation generator.

$$\begin{aligned}
\left[P_\mu, \varphi_\alpha(x)\right] &= i\partial_\mu\varphi_\alpha(x) \\
\left[D, \varphi_\alpha(x)\right] &= i\left(\Delta + x^\mu\partial_\mu\right)\varphi_\alpha(x) \\
\left[L_{\mu\nu}, \varphi_\alpha(x)\right] &= -i\left(x_\mu\partial_\nu - x_\nu\partial_\mu\right)\varphi_\alpha(x) + i\left(S_{\mu\nu}\right)_{\alpha\beta}\varphi_\beta(x) \\
\left[K_\mu, \varphi_\alpha(x)\right] &= 2ix_\mu\Delta\varphi_\alpha(x) + i\left(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\right)\varphi_\alpha(x) + 2ix^\rho\left(S_{\rho\mu}\right)_{\alpha\beta}\varphi_\beta(x)
\end{aligned} \tag{76}$$

One can also derive the finite transformation of the primary field, for example under the dilation transformation, we have:

$$\varphi'(\lambda x) = e^{i\lambda\hat{D}}\varphi(\lambda x)e^{-i\lambda\hat{D}} = \lambda^{-\Delta}\varphi(x). \tag{77}$$

For a scalar primary field, the finite transformation under a general conformal transformation is:

$$\varphi'(x') = \Omega(x)^{-\frac{\Delta}{d}}\varphi(x). \quad \Omega(x) = \det\frac{\partial x'^\mu}{\partial x^\nu} \tag{78}$$

For a generic scaling field that commute as:

$$[D, \varphi(0)] = i\Delta\varphi(0). \tag{79}$$

The algebra shows that P_μ raises the dimension by one, while K_μ lowers it by one:

$$\begin{aligned}\big[D, P_\mu\varphi(0)\big] &= i(\Delta+1)P_\mu\varphi(0),\\ \big[D, K_\mu\varphi(0)\big] &= i(\Delta-1)K_\mu\varphi(0).\end{aligned} \tag{80}$$

In a unitary theory the spectrum of dimensions is bounded below. Thus we can see that by acting the P_μ operators on a primary field we can generate a seires of scaling field with highier scaling dimensions:

$$\big[P_{\mu_1}...P_{\mu_n}, \varphi(0)\big] \sim \partial_{\mu_1}...\partial_{\mu_n}\varphi(0). \tag{81}$$

We call these fields descendants of the primary field.

2.3 Noether's theorem and the stress tensor

For an action

$$S = \int d^dx\mathcal{L}\big(\varphi, \partial_\mu\varphi\big), \tag{82}$$

Noether's theorem associates a conserved current to every continuous symmetry. In a general infinitesimal transformation, one convenient form is

$$j^\mu_a = \left[\frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\partial_\nu\varphi - \delta^\mu_\nu\mathcal{L}\right]\frac{\partial x^\nu}{\partial\omega_a} - \frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\frac{\partial\tilde{F}}{\partial\omega_a}, \qquad \partial_\mu j^\mu_a = 0. \tag{83}$$

The translation current is the canonical stress tensor,

$$T^{\mu\nu}_{\mathrm{c}} = -\eta^{\mu\nu}\mathcal{L} + \frac{\partial\mathcal{L}}{\partial\big(\partial_\mu\varphi\big)}\partial^\nu\varphi, \qquad \partial_\mu T^{\mu\nu}_{\mathrm{c}} = 0. \tag{84}$$

By an improvement one may choose a specific symmetric tensor $T^{\mu\nu} = T^{\nu\mu}$. Under a general coordinate transformation $x'^\mu = x^\mu + \varepsilon^\mu(x)$, we have:

$$\delta S = -\frac{1}{2}\int d^dx T^{\mu\nu}\big(\partial_\mu\varepsilon_\nu + \partial_\nu\varepsilon_\mu\big) \tag{85}$$

If we choose this general transformation to be a conformal transformation, we have due to the conformal killing equation:

$$\delta S = -\frac{1}{2}\int d^dx f(x)T^\mu{}_\mu \tag{86}$$

Scale invariance means that $\delta S = 0$ when $f(x) = \mathrm{const}$ this will lead to:

$$T^\mu{}_\mu = \partial_\mu V^\mu, \tag{87}$$

with a virial current V^μ. If we also have conformal invariance, we have $\delta S = 0$ for SCT, then:

$$T^\mu{}_\mu = 0. \tag{88}$$

In summary, in CFT we always can have a improved symmetric and traceless stress tensor $T^{\mu\nu}$, which is conserved.

2.4 Implications for correlation functions

For a QFT we define the correlation function as:

$$\langle\mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle = \left(\frac{1}{Z}\right)\int D\varphi\mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)e^{-S[\varphi]}, \tag{89}$$

notice that we always work in the Euclidean signature. One can prove that for a symmetry we have the result that:

$$\langle\varphi(x_1')...\varphi(x_n')\rangle = \langle\mathcal{F}(\varphi(x_1))...\mathcal{F}(\varphi(x_n))\rangle \tag{90}$$

By $\mathcal{F}(\varphi(x))$ we means the RHS of the definition of field condiguration transformation:

$$\varphi'(x') = \mathcal{F}(\varphi(x)). \tag{91}$$

which is the transformation purely of field (not the spacetime point). Notice that this is pure path integral and have nothing to do with whether the transformation is Unitary in certain time choice or not.

2.4.1 Two-point functions

Translation and rotation invariance imply dependence only on $x_{12} = x_1 - x_2$. Dilatations give

$$\langle\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\rangle = \frac{C_{12}}{(|x_{12}|)^{\Delta_1+\Delta_2}}. \tag{92}$$

Under an SCT,

$$x_{12'}^2 = \frac{x_{12}^2}{\gamma_1\gamma_2}, \qquad \gamma_i = 1 - 2b\cdot x_i + b^2 x_i^2. \tag{93}$$

SCT covariance requires $\Delta_1 = \Delta_2$. In an orthonormal basis of scalar primaries,

$$\langle\mathcal{O}_i(x_1)\mathcal{O}_j(x_2)\rangle = \frac{C_i\delta_{ij}}{(x_{12}^2)^{\Delta_i}}. \tag{94}$$

2.4.2 Three-point functions

For scalar primaries, conformal symmetry fixes the position dependence completely:

$$\langle\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = \frac{\lambda_{123}}{\left[(x_{12}^2)^{\frac{\Delta_1+\Delta_2-\Delta_3}{2}}(x_{13}^2)^{\frac{\Delta_1+\Delta_3-\Delta_2}{2}}(x_{23}^2)^{\frac{\Delta_2+\Delta_3-\Delta_1}{2}}\right]}. \tag{95}$$

2.4.3 Four-point functions and cross-ratios

For four identical scalar primaries of dimension Δ,

$$\langle\mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \frac{1}{\left[(x_{12}^2)^{\Delta}(x_{34}^2)^{\Delta}\right]}, \mathcal{G}(u,v), \tag{96}$$

where

$$u = \frac{x_{12}^2x_{34}^2}{x_{13}^2x_{24}^2} = z\overline{z}, \qquad v = \frac{x_{14}^2x_{23}^2}{x_{13}^2x_{24}^2} = (1-z)(1-\overline{z}). \tag{97}$$

Notice that the definition of z and $\overline{z}$ is conventionally not the same for Euclidean and Lorentzian theory (though the corrator will always take the same form). One can see that in Euclidean signature $\overline{z} = z^*$. In Lorentzian signature z and $\overline{z}$ are independent real variables in the standard kinematic region due to the contraction definition difference.

A standard Lorentzian configuration is

$$x_1 = 0, \qquad x_2 = (t, x, 0, ...), \qquad x_3 = (0, 1, 0, ...), \qquad x_4 = \infty. \tag{98}$$

One should be careful with what we mean by $x_4 = \infty$. In fact, we can take a limit of x_4 to infinity, and the correlation function will be defined as:

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \lim_{x_4\to\infty} (x_4^2)^{\Delta} \langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle. \tag{99}$$

Then

$$u = x^2 - t^2, \qquad v = (1-x)^2 - t^2, \qquad z = x - t, \qquad \overline{z} = x + t. \tag{100}$$

We can explicitly see that in the case z and $\overline{z}$ are independent real variables.

3 Radial Quantization

In the last discussion, we focus on a general notation that is better suited for lorentzian signature. In the Euclidean theory (especially in Radial quantization we will discuss), the Unitarity of the operators is not satisfied. A corresponding concept is the reflection positivity, one can see more on this discussion in Simmons-Duffin's lecture. A brief reason is that we want the wick rotation to be consistent, and the Lorentzian theory is unitary, the Euclidean theory will be reflection positive yet the Euclidean transformation is not unitarily represented.

However, we still want in the certain Hilbert space the time generators is Hermitian. Thus, we conventionally define the generators in another exponential map:

$$O = e^{-i\omega_a G_a} = e^{\omega_a G_a^E}, \tag{101}$$

$$P_\mu{}^E = -iP_\mu, \quad K_\mu{}^E = -iK_\mu, \quad M_{\mu\nu}{}^E = -iM_{\mu\nu}, \quad D^E = -iD. \tag{102}$$

correspondingly, the generators O will be:

$$\omega_a G_a^E = a_\mu P_\mu{}^E + b_\mu K_\mu{}^E + \frac{1}{2}\theta_{\mu\nu} M_{\mu\nu}{}^E + \lambda D^E. \tag{103}$$

We will see that the dilatation generators will be Hermitian in the radial quantization Hilbert space. From now on we will drop the superscript E and use the new convention for the generators. And the commutation relations between the generators and the local primaries will be changed into:

$$[D, \varphi_\alpha(0)] = \Delta\varphi_\alpha(0), \quad \big[M_{\mu\nu}, \varphi_\alpha(0)\big] = \big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(0), \quad \big[K_\mu, \varphi_\alpha(0)\big] = 0. \tag{104}$$

The action of the conformal generators at a general spacetime point can be derived using translations:

$$\begin{aligned}
\big[P_\mu, \varphi_\alpha(x)\big] &= \partial_\mu\varphi_\alpha(x), \\
[D, \varphi_\alpha(x)] &= \big(\Delta + x^\mu\partial_\mu\big)\varphi_\alpha(x), \\
\big[M_{\mu\nu}, \varphi_\alpha(x)\big] &= -\big(x_\mu\partial_\nu - x_\nu\partial_\mu\big)\varphi_\alpha(x) + \big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(x), \\
\big[K_\mu, \varphi_\alpha(x)\big] &= 2x_\mu\Delta\varphi_\alpha(x) + \big(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu\big)\varphi_\alpha(x) + 2x^\rho\big(S_{\mu\nu}\big)_{\alpha\beta}\varphi_\beta(x).
\end{aligned} \tag{105}$$

In this convention, the Euclidean conformal algebra is

$$\begin{aligned}
\big[D, P_\mu\big] &= P_\mu, \quad \big[D, K_\mu\big] = -K_\mu, \\
\big[K_\mu, P_\nu\big] &= 2\big(\delta_{\mu\nu}D - M_{\mu\nu}\big), \quad \big[M_{\mu\nu}, P_\rho\big] = -\big(\delta_{\mu\rho}P_\nu - \delta_{\nu\rho}P_\mu\big), \\
\big[M_{\mu\nu}, K_\rho\big] &= -\big(\delta_{\mu\rho}K_\nu - \delta_{\nu\rho}K_\mu\big), \\
\big[M_{\mu\nu}, M_{\rho\sigma}\big] &= -\delta_{\mu\rho}M_{\nu\sigma} + \delta_{\mu\sigma}M_{\nu\rho} + \delta_{\nu\rho}M_{\mu\sigma} - \delta_{\nu\sigma}M_{\mu\rho}, \\
\big[P_\mu, P_\nu\big] &= 0, \quad \big[K_\mu, K_\nu\big] = 0, \quad [D, D] = 0, \quad \big[D, M_{\mu\nu}\big] = 0.
\end{aligned} \tag{106}$$

Translations reconstruct an operator away from the origin is given by:

$$\mathcal{O}(x) = e^{x^\mu P_\mu}\mathcal{O}(0)e^{-x^\mu P_\mu}. \tag{107}$$

In this convention the descendants are defined as:

$$P_{\mu_1}...P_{\mu_n}\varphi_\alpha(0). \tag{108}$$

3.1 Radial Quantization

3.1.1 Lorentzian Theory in Cylinder Coordinate

Write a point in $\mathbb{R}^d$ as the following coordinates (τ, n):

$$x^\mu = rn^\mu, \quad r = e^\tau, \quad n^\mu n_\mu = 1. \tag{109}$$

One can verify that this is indeed a conformal transformation. For a **scalar primary** of dimension Δ, we define its form on the cylinder coordinates as:

$$\mathcal{O}_{\text{cyl}}(\tau, n) = e^{\tau\Delta}\mathcal{O}_{\text{flat}}(e^\tau, n). \tag{110}$$

The origin of the plane corresponds to $\tau \to -\infty$, while spatial infinity corresponds to $\tau \to +\infty$. From this relation, we can see how the conformal generators act on the cylinder coordinates. The commutation relations between the conformal generators and the cylinder coordinate primaries are given by:

$$\nabla_\mu{}^{\text{S}} := \left(\delta_\mu{}^\nu - n_\mu n^\nu\right)\frac{\partial}{\partial n^\nu}. \tag{111}$$

The conformal generators act on the cylinder operator as

$$\begin{aligned}
\left[D, \mathcal{O}_{\text{cyl}}(\tau, n)\right] &= \partial_\tau \mathcal{O}_{\text{cyl}}(\tau, n), \\
\left[M_{\mu\nu}, \mathcal{O}_{\text{cyl}}(\tau, n)\right] &= -\left(n_\mu \nabla_\nu{}^{\text{S}} - n_\nu \nabla_\mu{}^{\text{S}}\right)\mathcal{O}_{\text{cyl}}(\tau, n), \\
\left[P_\mu, \mathcal{O}_{\text{cyl}}(\tau, n)\right] &= e^{-\tau}\left(n_\mu(\partial^\tau - \Delta) + \nabla_\mu{}^{\text{S}}\right)\mathcal{O}_{\text{cyl}}(\tau, n), \\
\left[K_\mu, \mathcal{O}_{\text{cyl}}(\tau, n)\right] &= e^{\tau}\left(n_\mu(\partial^\tau + \Delta) - \nabla_\mu{}^{\text{S}}\right)\mathcal{O}_{\text{cyl}}(\tau, n).
\end{aligned} \tag{112}$$

We care about the cylinder coordinate because for a CFT in canonical formalism, we usually consider a Lorentzian CFT on a cylinder, we do the canonical quantization for lorentzian theory to get the Hilbert space. By wick rotation and coordinate symmetry transformation, we can investigate this theory in Euclidean cylinder, and with a coordinate transformation above we can have a theory on the plane.

> **Remark**
>
> Notice that we only do quantization in the lorentzian theory, and the Hilbert space sstays the same no matter what we do to the cooridnate transformation, its just symmetry transformations on the same Hilbert space.

From this procedure, the canonical Hamiltonian of the lorentzian theory (which is definitely a Hermite operator in the Hilbert space) which we quantized on will be something like:

$$U(t) = e^{-itH}, \quad H^\dagger = H. \quad \varphi(t, x) = U(t)^\dagger \varphi(0, x) U(t). \tag{113}$$

under the wick rotation, we have $t \to -i\tau$, and the Euclidean time evolution operator will be:

$$U(\tau) = e^{-\tau H}, \quad H^\dagger = H. \quad \varphi(\tau, x) = e^{\tau H}\varphi(0, x)e^{-\tau H}. \tag{114}$$

Notice that here the Hamiltonian is still Hermitian in the Hilbert space, but the Eculidean time evolution operator is not unitary. Taking a derivative on both side, this gives the commutation relation between the Hamiltonian and the local operators:

$$[H, \varphi(\tau, x)] = \partial_\tau \varphi(\tau, x). \tag{115}$$

One may notice that this is exactly what we called the dilatation operator D. Thus, we identify the dilatation operator (which is a abstract generator) with the Hamiltonian of the lorentzian theory on the cylinder, which is a Hermitian operator on the Hilbert space.

The rotation on the equal time slice in the original lorentzian theory will be generated by:

$$U(R) = e^{-i\frac{1}{2}\theta M^C_{\mu\nu}}, \quad \left(M^C_{\mu\nu}\right)^\dagger = M^C_{\mu\nu}. \tag{116}$$

However in our convention, the generator of the rotation is given by :

$$U(R) = e^{\frac{1}{2}\theta M_{\mu\nu}}, \quad M^\dagger_{\mu\nu} = -M_{\mu\nu}. \tag{117}$$

Thus, one may identify that

$$M_{\mu\nu} = -iM^C_{\mu\nu}. \tag{118}$$

However, due to we insist that the lorentzian theory is Unitary, the rotation generator is anti-Hermite in the Hilbert space.

3.1.2 Radial Quantization

Now we are convinced that it is consistent to define a canonical formalism of the CFT on the Euclidean plane as the one that quantized on the lorentzian theory on the Cylinder (which is well defined in ordinary QFT courses). However, it is too complicated to first do a quantization on a cylinder in lorentzian signature and then wick rotate to the Euclidean plane. Thus, what we do is that we develop a formalism purely on the plane but gives the exactly same Hilbert space as the Unitary lorentzian theory on the cylinder. This approach is called the **radial quantization**.

> **Remark**
>
> One may wonder doesn't that mean that the analytical continuation to a lorentzian CFT is uniquely fixed by the Euclidean time choice? Indeed, different choice of a Euclidean time in fact is correlated with different analytical continuation to a lorentzian CFT. Radial quantization is correspond to the analytical continuation of the cylinder coordinate. In one naively try to do an analytical continuation of the x^0 coordinate, the result won't be unitary therefore won't be a consistent CFT.
>
> This note only focus on the radial quantization and the consequently Euclidean theory. And we will not discuss lorentzian CFT.

Currently, I would understand the radial quantization is a canonical formalism of CFT that gives a Unitary theory in the cylinder coordinate in lorentzian time.

We take the following conditions for a Unitary Hilbert space:

- D is a Hermite Operator in the Hilbert space and is bounded from below.
- We exist a vacuum state $|0\rangle$ which is invariant under the conformal group.

For a state $|\psi\rangle$ in the Hilbert space, we can act n times with K_μ on it and due to D is bounded from below, we will eventually get a series of state $|\psi_0\rangle$ which is annihilated by all K_μ. Thus the Hilbert space should be spanned by the states of the form:

$$P_{\mu_1}...P_{\mu_n}|\psi_0\rangle. \tag{119}$$

We then can conclude the Hilbert space should be spanned by a series of primary states and its descendants. The primary states satisfies the following conditions:

$$K_\mu|\psi_\alpha\rangle = 0, \quad D|\psi_\alpha\rangle = \Delta|\psi_\alpha\rangle, \quad M_{\mu\nu}|\psi_\alpha\rangle = \left(S_{\mu\nu}\right)_{\alpha\beta}|\psi_\beta\rangle. \tag{120}$$

we then define a primary operator in this Hilbert space as:

Definition 3.1 **Primary Operator**

A primary operator correpsond to the state $|O_i\rangle$ is defined as a local operator O_i such that:

$$O_i(0)|0\rangle = |O_i\rangle, \tag{121}$$

Remark

This relation is more of a axiom that such operator always exists not merely if it exist we can do such identification.

Then one can find that these operator indeed have the correct commutation relation with the conformal generators as the ordinary conformal primary fields acting on the vaccum state at least (A natural generalization can be the commutation holds everywhere). More importantly, this construction tells us that a CFT can have a one-to-one correspondence between the primary operators and the primary states in the Hilbert space. This is called the **state-operator correspondence**.

Remark

One may encounter the statement that "state lives on a equal time slice". However, I don't think this statement is precise enough. I think what is truly means is that the Hilbert space is dependent on the choice of time coordinate. Different choice of time will lead to different inner product and all sort of structures.

YL: [The discussion here should not be understand as a prove or something, but an argument of universal structure. A rigourous treatment should be done in axiomatic CFTs.]

3.1.3 Correlation Functions

Before we only define the correlation function in the path integral formalism. Now with a canonical formalism, we need to find an equivalent way to compute the same object. We find out that the correlation function can be computed in the Hilbert space as:

$$\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle = \langle 0|\mathcal{R}\mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)|0\rangle. \tag{122}$$

The LHS is the correlation function defined in the path integral formalism, and the RHS is a correlator in the Hilbert space. (Of course, they should consistent with the same action S)

YL: [This certainly needs proof. However, one can convince themselves this is true.]

From now on, we will always use the simple notation of

$$\langle \mathcal{O}_1(x_1)...\mathcal{O}_n(x_n)\rangle \tag{123}$$

for the correlation function. However, the formalism we use may interchange between the path integral formalism and the canonical formalism due to the above equivalence.

3.1.4 What is "time" dependent

YL: [Discussion here is personal, I believe it needs further clarification.]

- Path Integral Formalism

In the path integral formalism, in Euclidean formalism, nothing is time dependent. The Euclidean path integral and correlation functions are only theory dependent unique object.

If we want to get a lorentzian theory from the Euclidean theory. We need to specify an analytical continuation. The choice of analytical continuation should depend on what we call "time" direction.

- Canonical Formalism

In the canonical formalism, in Euclidean formalism. The Hilbert space depends on what we called an Euclidean time direction. However, all choice of time should give us the exact same correlation functions that matches the path integral formalism.

If we want to get a lorentzian theory from a Euclidean theory, we only get the one that the Euclidean time specified us for.

3.1.5 Radial Conjugation

For a scalar primary, a consistent radial conjugation (The complex conjugation of radial quantization) acts as inversion:

$$(\mathcal{O}(x))^\dagger = \frac{1}{|x|^{2\Delta}}\mathcal{O}\Big(\frac{x}{x^2}\Big). \tag{124}$$

The bra corresponding to $|\mathcal{O}\rangle$ is therefore

$$\langle\mathcal{O}| = \lim_{|x|\to\infty} |x|^{2\Delta}\langle 0|\mathcal{O}(x). \tag{125}$$

If the two-point function is normalized by

$$\langle\mathcal{O}(x)\mathcal{O}(0)\rangle = \frac{1}{|x|^{2\Delta}}, \tag{126}$$

then

$$\langle\mathcal{O}|\mathcal{O}\rangle = \lim_{|x|\to\infty} |x|^{2\Delta}\langle\mathcal{O}(x)\mathcal{O}(0)\rangle = 1. \tag{127}$$

The adjoint relations between conformal generators within this Hermite conjugation are:

$$D^\dagger = D, \quad {P_\mu}^\dagger = K_\mu \quad D^\dagger = D, \quad M_{\mu\nu}^\dagger = -M_{\mu\nu}. \tag{128}$$

These relations are the key input in the derivation of the unitarity bounds below.

3.2 Operator product expansion

Consider two local operators φ_1 and φ_2 inserted at points x and 0. In radial quantization, the state created by these two operators is:

$$\varphi_1(x)\varphi_2(0)|0\rangle. \tag{129}$$

Expanding this state in a complete basis of dilatation eigenstates gives a convergent sum over conformal multiplets:

$$\varphi_1(x)\varphi_2(0)|0\rangle = \sum_i C_i(x,\partial)\mathcal{O}_i(0)|0\rangle. \tag{130}$$

Equivalently, inside correlation functions, we can formally write a relation:

$$\varphi_1(x)\varphi_2(0) = \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} C_{\mathcal{O}}(x,\partial)\mathcal{O}(0). \tag{131}$$

The sum runs over primary operators; the differential operator $C_{\mathcal{O}}(x,\partial)$ generates the complete tower of descendants. Conformal symmetry fixes this differential operator in terms of the dimensions and spins, up to the theory-dependent OPE coefficient $\lambda_{12\mathcal{O}}$.

> **Remark**
>
> In fact, a general OPE should be defined within the Radial ordering of the operators.
>
> $$\mathcal{R}\varphi_1(x)\varphi_2(y) = \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} C_{\mathcal{O}}(x,y,\partial)\mathcal{O}(y). \tag{132}$$
>
> Here it is implicit cause we choose x and 0 points. Field at different radius can't generally assumed to commute, as we care mostly on correlators, we defined OPE by the radial ordering, whose action on the vaccum directly give the correlator.

Due to the identification of correlation functions in the path integral formalism and the canonical formalism, we can also define OPE equivalently in the path integral formalism:

$$\varphi_1(x)\varphi_2(0) = \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} C_{\mathcal{O}}(x,\partial)\mathcal{O}(0). \tag{133}$$

here we don't have operators and states (and of course radial ordering). This expansion make sense only insider the correlation function.

3.2.1 Leading terms and the first descendant

For a scalar primary $\mathcal{O}$ of dimension Δ appearing in the OPE of scalar primaries φ_1 and φ_2, the leading terms are

$$\varphi_1(x)\varphi_2(0) \supseteq \frac{\lambda_{12\mathcal{O}}}{|x|^k}\big(\mathcal{O}(0) + \alpha x^\mu \partial_\mu \mathcal{O}(0) + ...\big). \tag{134}$$

We can match the coefficients by the following approach. We compare the action of D, K_μ on the OPE. However, one way is that we first do an OPE expand to the RHS and then act with the generators. The other way is that we first act with the generators on the LHS and then do an OPE expand to the RHS. The two results should be the same. Comparing the terms should fix the coefficients.

Acting with D fixes the power of $|x|$:

$$k = \Delta_1 + \Delta_2 - \Delta. \tag{135}$$

Acting with the special conformal generator fixes the relative coefficient of the first descendant. For a scalar primary,

$$\big[K_\mu, \mathcal{O}(x)\big] = \big(2\Delta x_\mu + 2x_\mu x^\nu \partial_\nu - x^2\partial_\mu\big)\mathcal{O}(x). \tag{136}$$

The result is

$$\alpha = \frac{\Delta_1 + \Delta - \Delta_2}{2\Delta}. \tag{137}$$

In particular, the numerator contains $\Delta_1 - \Delta_2$, not $\Delta_1 + \Delta_2$. The identity contribution $\Delta = 0$ must be treated separately and has no descendants.

The same coefficient can be obtained directly by inserting the OPE into a three-point function. Conformal invariance fixes

$$\langle \varphi_1(x_1)\varphi_2(x_2)\mathcal{O}(x_3)\rangle = \frac{\lambda_{12\mathcal{O}}}{|x_{12}|^{\Delta_1+\Delta_2-\Delta}|x_{23}|^{\Delta_2+\Delta-\Delta_1}|x_{13}|^{\Delta_1+\Delta-\Delta_2}}. \tag{138}$$

And pluging in the OPE ansatz into the 3 point correlation function for the first two fields and then use the equivalent of the canonical and path integral formalism tells us that :

$$\begin{aligned}\langle\varphi_1(x)\varphi_2(0)\mathcal{O}(x_3)\rangle &= \langle 0|\mathcal{R}\varphi_1(x)\varphi_2(0)\mathcal{O}(x_3)|0\rangle \\ &= \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} C_{\mathcal{O}}\left(x_{12},\partial_{x_2}\right)\langle\mathcal{O}(x_2)\mathcal{O}(x_3)\rangle|_{x_2=0}.\end{aligned} \tag{139}$$

3.3 Unitarity bounds

In radial quantization, positivity of descendant norms constrains the possible scaling dimensions. For a scalar primary state $|\mathcal{O}\rangle$,

$$\left\langle\mathcal{O}|K_\mu P_\nu|\mathcal{O}\right\rangle = 2\Delta\delta_{\mu\nu}\langle\mathcal{O}|\mathcal{O}\rangle. \tag{140}$$

Thus positivity at level one implies $\Delta \geq 0$. The stronger scalar bound comes from the level-two descendant

$$P^\mu P_\mu|\mathcal{O}\rangle. \tag{141}$$

Its norm is

$$\left\|P^\mu P_\mu|\mathcal{O}\rangle\right\|^2 = 8d\Delta\left(\Delta - \frac{d-2}{2}\right)\||\mathcal{O}\rangle\|^2. \tag{142}$$

Therefore a scalar primary in a unitary CFT satisfies

$$\Delta = 0 \quad \text{for the identity,} \quad \text{or} \quad \Delta \geq \frac{d-2}{2}. \tag{143}$$

For a symmetric traceless primary of spin $\ell \geq 1$,

$$\mathcal{O}_{\mu_1\ldots\mu_\ell}(x), \tag{144}$$

positivity of the level-one divergence descendant gives

$$\Delta \geq d - 2 + \ell. \tag{145}$$

3.4 Conformal Blocks

For four identical scalar primaries of dimension Δ_φ, conformal symmetry gives

$$\langle\varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4)\rangle = \frac{\mathcal{G}(u,v)}{|x_{12}|^{2\Delta_\varphi}|x_{34}|^{2\Delta_\varphi}}, \tag{146}$$

where the conformal cross-ratios are

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \quad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}. \tag{147}$$

Applying the OPE to the pairs (12) and (34) yields the conformal-block decomposition

$$\mathcal{G}(u,v) = \sum_{\mathcal{O}} \lambda^2_{\varphi\varphi\mathcal{O}} G_{\Delta,\ell}(u,v). \tag{148}$$

More explicitly, we have:

$$\left[C_{\Delta(x_{12},\partial_y)} C_{\Delta(x_{34},\partial_z)}\langle\varphi_\Delta(y)\varphi_\Delta(z)\rangle\right]\ |_{y=\frac{x_1+x_2}{2},z=\frac{x_3+x_4}{2}} = \frac{G_{\Delta,\ell}(u,v)}{|x_{12}|^{2\Delta_E}|x_{34}|^{2\Delta_E}} \tag{149}$$

The conformal block $G_{\Delta,\ell}(u,v)$ contains the contribution of one exchanged primary of dimension Δ and spin ℓ, together with all of its descendants. Once the external dimensions and the

quantum numbers (Δ, ℓ) are specified, the block is fixed entirely by conformal symmetry.It solves the Casimir differential equation with appropriate boundary conditions.

4 2D Conformal Field Theory

4.1 2D Conformal Algebra

Recall the conformal Killing equation for $x'^{\mu} = x^{\mu} + \varepsilon^{\mu}(x)$:

$$\partial_{\mu}\varepsilon_{\nu} + \partial_{\nu}\varepsilon_{\mu} = \frac{2}{d}(\partial_{\rho}\varepsilon^{\rho})\eta_{\mu\nu}. \tag{150}$$

In two Euclidean dimensions this gives

$$\partial_0\varepsilon_0 = \partial_1\varepsilon_1, \quad \partial_0\varepsilon_1 = -\partial_1\varepsilon_0. \tag{151}$$

Let

$$z = x^0 + ix^1, \quad \overline{z} = x^0 - ix^1, \tag{152}$$

$$\partial = \partial_z = \frac{1}{2}(\partial_0 - i\partial_1), \quad \overline{\partial} = \partial_{\overline{z}} = \frac{1}{2}(\partial_0 + i\partial_1). \tag{153}$$

The nonzero metric components are

$$g_{z\overline{z}} = g_{\overline{z}z} = \frac{1}{2}, \quad g^{z\overline{z}} = g^{\overline{z}z} = 2. \tag{154}$$

Writing $\varepsilon = \varepsilon^0 + i\varepsilon^1$ and $\overline{\varepsilon} = \varepsilon^0 - i\varepsilon^1$, we obtain

$$\overline{\partial}\varepsilon = 0, \quad \partial\overline{\varepsilon} = 0. \tag{155}$$

Hence $\varepsilon = \varepsilon(z)$ is holomorphic and $\overline{\varepsilon} = \overline{\varepsilon}(\overline{z})$ is antiholomorphic. Finite conformal coordinate transformations have the form

$$z' = f(z), \quad \overline{z}' = \overline{f}(\overline{z}). \tag{156}$$

Expand the infinitesimal transformation as

$$\varepsilon(z) = \sum_n c_n z^{n+1}, \quad \overline{\varepsilon}(\overline{z}) = \sum_n \overline{c}_n \overline{z}^{n+1}. \tag{157}$$

For a scalar function, the induced variation at fixed coordinates is

$$\delta_{\varepsilon}F(z,\overline{z}) = F(z - \varepsilon(z), \overline{z} - \overline{\varepsilon}(\overline{z})) - F(z,\overline{z}) = -\varepsilon(z)\partial F - \overline{\varepsilon}(\overline{z})\overline{\partial}F. \tag{158}$$

Thus we introduce the generators

$$\ell_n = -z^{n+1}\partial, \quad \overline{\ell}_n = -\overline{z}^{n+1}\overline{\partial}. \tag{159}$$

They form two commuting copies of the Witt algebra:

$$[\ell_m, \ell_n] = (m-n)\ell_{m+n}, \quad \left[\overline{\ell}_m, \overline{\ell}_n\right] = (m-n)\overline{\ell}_{m+n}, \quad \left[\ell_m, \overline{\ell}_n\right] = 0. \tag{160}$$

Consider the Riemann sphere S^2, with coordinate charts z and $z' = -\frac{1}{z}$. The vector field ℓ_n is regular at $z = 0$ only if $n \geq -1$, and regular at $z' = 0$ only if $n \leq 1$. Hence the globally defined holomorphic generators are $\ell_{-1}, \ell_0, \ell_1$, together with their antiholomorphic counterparts. They form the global conformal algebra. The combination $\ell_0 + \overline{\ell}_0$ generates dilatations, while $i\big(\ell_0 - \overline{\ell}_0\big)$ generates rotations.

Finite global transformations are Möbius maps:

$$z \to \frac{az+b}{cz+d}, \quad ad - bc = 1. \tag{161}$$

Their group is $PSL(2,\mathbb{C}) = \frac{SL(2,\mathbb{C})}{\{+1,-1\}} \cong SO^+(3,1)$.

4.2 Primaries and EM Tensor

Define a two-dimensional conformal primary by its transformation under every local conformal map:

Definition 4.1 **Primary Field**

A primary field $\varphi(z,\overline{z})$ of weights $\left(h,\overline{h}\right)$ transforms as:

$$\varphi'(z',\overline{z}')=\left(\frac{\partial z'}{\partial z}\right)^{-h}\left(\frac{\partial \overline{z}'}{\partial \overline{z}}\right)^{-\overline{h}}\varphi(z,\overline{z}). \tag{162}$$

A quasiprimary obeys this law under global conformal transformations only. Its scaling dimension and spin are defined as:

$$\Delta=h+\overline{h},\quad s=h-\overline{h}. \tag{163}$$

Remark

Notice that here we are not only dealing with scalar primaries, but also fields with spin. Due to we are working in 2D, spin are only represented by a single number and U(1) phase. In general the spin is only integer or half-integer.

For a primary field, the infinitesimal transformation law is:

$$\delta_{\varepsilon,\overline{\varepsilon}}\varphi(z,\overline{z})=-(h\partial\varepsilon+\varepsilon\partial)\varphi(z,\overline{z})-\left(\overline{h\partial}\overline{\varepsilon}+\overline{\varepsilon}\overline{\partial}\right)\varphi(z,\overline{z}). \tag{164}$$

Sometimes we separate the holomorphic and antiholomorphic parts, and give the notation of:

$$\delta_{\varepsilon}\varphi(z,\overline{z})=-(h\partial\varepsilon+\varepsilon\partial)\varphi(z,\overline{z}),\quad \delta_{\overline{\varepsilon}}\varphi(z,\overline{z})=-\left(\overline{h\partial}\overline{\varepsilon}+\overline{\varepsilon}\overline{\partial}\right)\varphi(z,\overline{z}). \tag{165}$$

However, one should remember that only the sum of the two variations is the actual infinitesimal conformal transformation of the field. For quasi-primary we have the same transformation under global conformal transformations, but not under local ones.

4.2.1 Holomorphic stress tensor

Symmetry and tracelessness imply

$$g^{\mu\nu}T_{\mu\nu}=2(T_{z\overline{z}}+T_{\overline{z}z})=0, \tag{166}$$

so $T_{z\overline{z}}=0$. Conservation then gives

$$\partial^{\mu}T_{\mu\nu}=0\quad\Rightarrow\quad\overline{\partial}T_{zz}=0,\quad\partial T_{\overline{z}\overline{z}}=0. \tag{167}$$

Define the holomorphic and antiholomorphic components by

$$T(z)=-2\pi T_{zz},\quad\overline{T}(\overline{z})=-2\pi T_{\overline{z}\overline{z}}. \tag{168}$$

4.3 Conformal correlators

Under global conformal transformations, quasi-primaries transform as the above way. Assuming such a symmetry gives the following restrictions on the correlators:

These formulas also hold for quasiprimaries. For primaries of weights $\left(h_i,\overline{h}_i\right)$,

$$\left\langle\varphi_i(z_1,\overline{z}_1)\varphi_j(z_2,\overline{z}_2)\right\rangle=\frac{C_{ij}}{z_{12}^{2h_i}\overline{z}_{12}^{2\overline{h}_i}}. \tag{169}$$

The coefficient can be nonzero only when the weights match. The three-point function is

$$\begin{aligned}&\left\langle\varphi_i(z_1,\overline{z}_1)\varphi_j(z_2,\overline{z}_2)\varphi_k(z_3,\overline{z}_3)\right\rangle\\&=\frac{\lambda_{ijk}}{z_{12}^{h_i+h_j-h_k}z_{13}^{h_i+h_k-h_j}z_{23}^{h_j+h_k-h_i}\overline{z}_{12}^{\overline{h}_i+\overline{h}_j-\overline{h}_k}\overline{z}_{13}^{\overline{h}_i+\overline{h}_k-\overline{h}_j}\overline{z}_{23}^{\overline{h}_j+\overline{h}_k-\overline{h}_i}}.\end{aligned}\tag{170}$$

Four-point functions are more complicated. For identical holomorphic fields, their coordinate dependence is a fixed prefactor times a function of the cross ratio

$$\eta=\frac{z_{12}z_{34}}{z_{13}z_{24}}.\tag{171}$$

A general form of the 4 point function is:

$$\langle\varphi(z_1,\overline{z}_1)\varphi(z_2,\overline{z}_2)\varphi(z_3,\overline{z}_3)\varphi(z_4,\overline{z}_4)\rangle=\frac{G(\eta,\overline{\eta})}{z_{12}^{2h}z_{34}^{2h}\overline{z}_{12}^{2\overline{h}}\overline{z}_{34}^{2\overline{h}}}.\tag{172}$$

where $G(\eta,\overline{\eta})$ is some function that only depends on the cross ratio.

4.4 Conformal Generators and the OPE

One may consistently find that the conformal transformations are generated by the following charges. In our conventions,

$$Q=\frac{1}{2\pi i}\oint\left(\mathrm{d}z\varepsilon(z)T(z)+\mathrm{d}\overline{z}\overline{\varepsilon}(\overline{z})\overline{T}(\overline{z})\right).\tag{173}$$

> **Remark**
>
> One should question themselves, what does the Symmetry Generator means. A concrete definition is that its commutation relation with an local operator generates the symmetry transformation. Though usually symmetry generator serves more that this.

For a holomorphic variation, one may find that:

$$\delta_\varepsilon\psi(w)=-\frac{1}{2\pi i}\left[\oint_C\mathrm{d}z\varepsilon(z)T(z),\psi(w)\right],\tag{174}$$

The countour should be a equal time circle of $|z|=|w|$.

> **Remark**
>
> One may question, why do we have this equality. This should be understood as a consequence of the Ward identity in the path integral formalism. In this note we choose to present it first and see that it indeed consist with the ward identity, which will not be proved in this note independently. A detailed and more rigorous discussion can be found in the yellow book.

For holomorphic $a(z)$ and another field $b(w)$, a contour just outside $|w|$ minus one just inside $|w|$ gives

$$\oint_{C_w}\mathrm{d}z\mathcal{R}a(z)b(w)=\oint_{C_+}\mathrm{d}za(z)b(w)-\oint_{C_-}\mathrm{d}zb(w)a(z)=\left[\oint_{|z|=|w|}\mathrm{d}za(z),b(w)\right].\tag{175}$$

Thus we get an identity that the commutator of the charge with a local field is equavlent to make a contour right around the field with radial ordering.

The infinitesimal transformation law of a primary is

$$\delta_{\varepsilon,\bar{\varepsilon}}\varphi(z,\bar{z}) = -(h\partial\varepsilon + \varepsilon\partial)\varphi(z,\bar{z}) - \left(\overline{h\partial}\bar{\varepsilon} + \bar{\varepsilon}\bar{\partial}\right)\varphi(z,\bar{z}). \tag{176}$$

Focus on the holomorphic part and write the singular expansion

$$\mathcal{R}[T(z)\psi(w)] \sim \sum_n (z-w)^n A_n(w). \tag{177}$$

Inserting it into the contour integral gives

$$\frac{1}{2\pi i}\oint_{C_w} \mathrm{d}z\varepsilon(z)\mathcal{R}[T(z)\psi(w)] = ... + \frac{\varepsilon''(w)A_{-3}(w)}{2} + \varepsilon'(w)A_{-2}(w) + \varepsilon(w)A_{-1}(w) + ... \tag{178}$$

Matching the primary transformation law, we find only two nonzero coefficients in the irregular part of the OPE:

$$A_{-2} = h\psi, \quad A_{-1} = \partial\psi. \tag{179}$$

> **Remark**
>
> From this relation we know nothing about the regular parts of the OPE, like $A_0, A_1, ...$ Cause the contour integral of these terms will vanish.

Therefore the singular OPE is

$$T(z)\psi(w) \sim \frac{h\psi(w)}{(z-w)^2} + \frac{\partial\psi(w)}{z-w}. \tag{180}$$

We use the notation $\sim$ to indicate that the equality holds only for the singular part of the OPE. The regular part is not fixed. (we will omit the radial ordering symbol in the OPE, However, one should always keep in mind that the OPE is a radial ordered product.)

More generally, an OPE can include local operators and their descendants. If O_k is a primary appearing in O_1O_2, its leading term has powers

$$O_1(z,\bar{z})O_2(w,\bar{w}) \sim \sum_k C_{12}^k (z-w)^{h_k-h_1-h_2}(\bar{z}-\bar{w})^{\bar{h}_k-\bar{h}_1-\bar{h}_2}O_k(w,\bar{w}) + ... \tag{181}$$

This is given by similar calculation as the highier dimensional case and the coefficient C_{12}^k is theory dependent. One may see that if we normalize the two point function coefficient to be one, then C_{12}^k is the same as the three point function coefficient λ_{12k}.

4.5 Example: free scalar

Now we illustrate the OPE of stress tensor with a primary field in a concrete example. Consider the free scalar action

$$S = \frac{g}{2}\int \mathrm{d}^2x\partial_\mu\varphi(x)\partial^\mu\varphi(x). \tag{182}$$

Its propagator in $d = 2$ is logarithmic:

$$\langle\varphi(x)\varphi(y)\rangle = -\frac{1}{4\pi g}\ln(x-y)^2 + \text{const.} \tag{183}$$

In terms of the holomrophic and antiholomorphic coordinates, we have:

$$\langle\varphi(z,\bar{z})\varphi(w,\bar{w})\rangle = -\frac{1}{4\pi g}\ln(z-w) - \frac{1}{4\pi g}\ln(\bar{z}-\bar{w}) + \text{const.} \tag{184}$$

The equation of motion is

$$\partial^\mu \partial_\mu \varphi = 0 \quad \Rightarrow \quad 4\partial\overline{\partial}\varphi = 0. \tag{185}$$

Locally, one can write $\varphi(z,\overline{z}) = \varphi(z) + \overline{\varphi}(\overline{z})$. We know that the propagator satisfies the EoM. Thus we can split the propagator. And as we see the holomorphic and anti-holomorphic part is separated. The holomorphic correlator is

$$\langle \varphi(z)\varphi(w)\rangle = -\frac{1}{4\pi g}\ln(z-w). \tag{186}$$

The logarithm means that φ itself is not an ordinary scaling primary, whereas

$$\langle \partial\varphi(z)\partial\varphi(w)\rangle = \partial_z\partial_w\left(-\frac{1}{4\pi g}\ln(z-w)\right) = -\frac{1}{4\pi g(z-w)^2}. \tag{187}$$

Thus $\partial\varphi$ is a candidate primary with weights $\left(h,\overline{h}\right) = (1,0)$. Check its OPE with $T(z)$. The classical canonical stress tensor is

$$T_{\mu\nu} = g\left(\partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}\eta_{\mu\nu}\partial_\rho\varphi\partial^\rho\varphi\right). \tag{188}$$

and in the holomorphic coordinates,

$$T(z) = -2\pi T_{zz} = -2\pi g\partial\varphi(z)\partial\varphi(z). \tag{189}$$

Now in the quantum theory, canonical quantization and path integral quantization both tells us that the quantum version of any composite operator should be defined with normal ordering.

$$T(z) = -2\pi g : \partial\varphi(z)\partial\varphi(z) : . \tag{190}$$

We would now focus more on path integrals, cause it is more convenient in calculating OPEs of free theories.

Normal ordering defines the coincident-point product in the path integral formalism (the following equality should be understood within the correlator):

$$T(z) = -2\pi g \lim_{w\to z}\left(\partial\varphi(z)\partial\varphi(w) + \frac{1}{4\pi g(z-w)^2}\right). \tag{191}$$

and in canonical formalism, an equivalent definition is putting the annihilation operators to the right of the creation operators.

Using Wick contractions and re-expanding around w gives

$$T(z)\partial\varphi(w) \sim \frac{\partial\varphi(w)}{(z-w)^2} + \frac{\partial^2\varphi(w)}{z-w}. \tag{192}$$

The double-pole coefficient is one, confirming that the $\partial\varphi$ field is a holomorphic primary operator with $h = 1$. We here also neglect the radial ordering symbol cause we're doing in path integral. However, when switching formalisms we should always keep the radial ordering in mind.

4.6 Conformal Ward identity

Let C enclose all insertions of primary fields of weight $\left(h_i,\overline{h}_i\right)$. Complex analysis tells us that this calculation can be deform it into small contours around the fields:

$$\begin{aligned}&\left\langle \frac{1}{2\pi i}\oint_C \mathrm{d}z\varepsilon(z)T(z)\varphi_1(w_1,\overline{w}_1)...\varphi_n(w_n,\overline{w}_n)\right\rangle \\ &=\sum_{j=1}^{n}\left\langle \varphi_1...\frac{1}{2\pi i}\oint_{C_{w_j}}\mathrm{d}z\varepsilon(z)T(z)\varphi_j\big(w_j,\overline{w}_j\big)...\varphi_n\right\rangle.\end{aligned} \tag{193}$$

The OPE at w_j yields the RHS to have the form:

$$\big(h_j\varepsilon'\big(w_j\big)+\varepsilon\big(w_j\big)\partial_{w_j}\big)\langle\varphi_1(w_1,\overline{w}_1)...\varphi_n(w_n,\overline{w}_n)\rangle. \tag{194}$$

With the variation convention above, this contribution equals $-\delta_\varepsilon\varphi_j$ inside the correlator. Since the contour relation holds for any $\varepsilon(z)$, the conformal Ward identity is

$$\left\langle T(z)\prod_{i=1}^{n}\varphi_i(w_i,\overline{w}_i)\right\rangle=\sum_{j=1}^{n}\left(\frac{h_j}{\big(z-w_j\big)^2}+\frac{1}{z-w_j}\partial_{w_j}\right)\left\langle\prod_{i=1}^{n}\varphi_i(w_i,\overline{w}_i)\right\rangle. \tag{195}$$

Regular OPE terms give zero on each small contour. The formula assumes the usual behaviour of $T(z)$ at infinity.

> **Remark**
>
> In fact the conformal Ward Identity is derived from the path integral formalism independently. Then an OPE between primaries and descendants can be derived from the Ward identity. Then the form of conformal generator follows, by defining they generate the conformal transformation. This discussion reverse the logic, though everything is consistent.

4.7 Transformation of $T(z)$

Is $T(z)$ a primary? No. It is a quasiprimary with weights $\big(h,\overline{h}\big)=(2,0)$. Its self-OPE is

$$T(z)T(w)\sim\frac{c}{2(z-w)^4}+\frac{2T(w)}{(z-w)^2}+\frac{\partial T(w)}{z-w}. \tag{196}$$

This OPE can be calculated by doing two conformal transformation in sequence.

Under a finite holomorphic change of coordinates $z'=f(z)$,

$$T'(z')=\left(\frac{\partial f}{\partial z}\right)^{-2}\Big(T(z)-\frac{c}{12}\{f,z\}\Big), \tag{197}$$

where the Schwarzian derivative is

$$\{f,z\}=\frac{f'''(z)}{f'(z)}-\frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2. \tag{198}$$

The central charge c represents a quantum effect and depends on the CFT. For a free boson, $c=1$.

5 Minimal Models

5.1 Virasoro algebra

Define the modes of the stress tensors by

$$L_n = \frac{1}{2\pi i}\oint \mathrm{d}z z^{n+1} T(z), \quad \overline{L}_n = \frac{1}{2\pi i}\oint \mathrm{d}\overline{z} \overline{z}^{n+1} \overline{T}(\overline{z}). \tag{199}$$

Then

$$T(z) = \sum_{n\in\mathbb{Z}} z^{-n-2} L_n. \tag{200}$$

For an infinitesimal transformation

$$\varepsilon(z) = \sum_n \varepsilon_n z^{n+1}, \tag{201}$$

its charge is

$$Q_\varepsilon = \sum_n \varepsilon_n L_n. \tag{202}$$

Let us compute $[L_n, L_m]$. Using two independent contours, we bring the z contour to a small circle C_w around w. Only the singular terms in the $T(z)T(w)$ OPE contribute:

$$[L_n, L_m] = \frac{1}{(2\pi i)^2}\oint \mathrm{d}w w^{m+1} \oint_{C_w} \mathrm{d}z z^{n+1} \frac{c}{2(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{203}$$

Taking the z residues gives

$$[L_n, L_m] = \frac{1}{2\pi i}\oint \mathrm{d}w w^{m+1} \frac{c}{12}(n+1)n(n-1)w^{n-2} + 2(n+1)w^n T(w) + w^{n+1}\partial T(w). \tag{204}$$

The first two terms yield

$$\frac{c}{12} n(n^2-1)\delta_{n+m,0} + 2(n+1)L_{n+m}. \tag{205}$$

For the last term, integration by parts gives

$$\frac{1}{2\pi i}\oint \mathrm{d}w w^{m+n+2}\partial T(w) = -(m+n+2)L_{n+m}. \tag{206}$$

Therefore

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{c}{12} n(n^2-1)\delta_{n+m,0}. \tag{207}$$

This is the **Virasoro algebra**: the Witt algebra with a central extension. The central term commutes with all generators and is related to quantum effects, such as arbitrary phases in wave functions.

Also one may check the commutation relation with primary fields:

$$[L_n, \varphi(w)] = \frac{1}{2\pi i}\oint_{C_w} \mathrm{d}z z^{n+1} T(z)\varphi(w) = h(n+1)w^n + w^{n+1}\partial\varphi(w). \tag{208}$$

5.2 Hilbert space of CFT

We question what is a universal behavior of a CFT Hilbert space. A CFT should form a representation (may not be unitary) of the Virasoro algebra. A general structure can be constructed as follows. We define **Virasoro primary** state obeys

$$L_n|\varphi\rangle = 0 \quad \text{for } n \geq 1, \qquad \overline{L}_n|\varphi\rangle = 0 \quad \text{for } n \geq 1, \tag{209}$$

and

$$L_0|\varphi\rangle = h|\varphi\rangle, \qquad \overline{L}_0|\varphi\rangle = \overline{h}|\varphi\rangle. \tag{210}$$

Acting with negative modes produces its Virasoro descendants. The resulting Virasoro multiplet is called a **Verma module**. For example,

$$|h\rangle, \quad L_{-1}|h\rangle, \quad L_{-1}^2|h\rangle, \quad L_{-2}|h\rangle, \tag{211}$$

and at the next level,

$$L_{-1}^3|h\rangle, \quad L_{-1}L_{-2}|h\rangle, \quad L_{-3}|h\rangle. \tag{212}$$

A mode L_{-n} raises the L_0 eigenvalue by n:

$$[L_0, L_{-n}] = nL_{-n}. \tag{213}$$

For special values of c and h, a linear combination $|\chi\rangle$ of descendants can itself satisfy the highest-weight conditions

$$L_n|\chi\rangle = 0 \quad \text{for every } n \geq 1. \tag{214}$$

5.2.1 Realization of Hilbert Space

In the above discussion, we only abstractly defined the Hilbert space of a CFT. We need a realization for the Verma module. We may find out that we only need to assume the existence of a EM tensor, a series of primary operators. Then the Hilbert space will be grouped into the Verma modules.

First it should have a conformal invariant unique vaccum. We assume a vaccum that satisfies:

$$L_n|0\rangle = 0 \quad \text{for } n \geq -1. \tag{215}$$

> **Remark**
>
> This condition in fact follows from the assumption that the EM tensor act on vaccum is well defined at 0 point.

This includes the global generators L_{-1}, L_0, L_1. In general,

$$L_{-2}|0\rangle \neq 0. \tag{216}$$

In fact, $L_{-2}|0\rangle$ is exactly the state $T(0)|0\rangle$.

We define primary state by acting primary operator on vaccum:

$$|\varphi\rangle = \varphi(0)|0\rangle. \tag{217}$$

Then the descendant states are defined by acting the negative modes of Virasoro algebra on the primary state:

$$L_{-n_1}...L_{-n_k}|\varphi\rangle \tag{218}$$

One may notice that the descendant states can also be defined by acting some operators on the vaccum state. They will be combination of the EM tensor and the primary operators.we define them as **descendent operator**:

$$\varphi^{-n}(w) = (L_{-n}\varphi)(w) = \frac{1}{2\pi i}\oint_{C_w} \mathrm{d}z (z-w)^{n-1} T(z)\varphi(w). \tag{219}$$

One may realize by definition we have:

$$L_{-n}|\varphi\rangle = \varphi^{-n}(0)|0\rangle. \tag{220}$$

We now see that we not only realized the Verma module, but also realized an operator for each state in the Verma module. This is the state-operator correspondence.

> **Remark**
>
> I think one shouldn't think of state-operator correspondence some incredible property but just some result of construction by hand, cause we realize the Hilbert space by acting operators on vaccum, there must be such a correspondence.

5.2.2 Complex Conjugation

In radial quantisation, conjugation relates a field at z to one at the inverted point $\frac{1}{\overline{z}}$; for a primary of weights $\left(h, \overline{h}\right)$,

$$\varphi^{\dagger(z,\overline{z})} = \frac{1}{z^{2h}\overline{z}^{2\overline{h}}}\varphi\left(\frac{1}{\overline{z}}, \frac{1}{z}\right). \tag{221}$$

> **Remark**
>
> This is similar to the highier dimensional case. We choose such conjugation to make it a Unitary CFT on a cylinder.

In particular, one may find out

$$L_n^\dagger = L_{-n}. \tag{222}$$

Consequently,

$$\langle h|L_n = 0 \quad \text{for } n \leq -1. \tag{223}$$

5.3 CFT Correlators

5.3.1 Correlators of descendants

Its correlators can be evaluated by inserting a contour integral:

$$\left\langle \varphi^{-n}(w)\varphi_1(w_1)...\varphi_{N(w_N)}\right\rangle = \frac{1}{2\pi i}\oint_{C_w} \frac{\mathrm{d}z}{(z-w)^{n-1}}\left\langle T(z)\varphi(w)\varphi_1(w_1)...\varphi_{N(w_N)}\right\rangle. \tag{224}$$

Deform the contour and use the conformal Ward identity. This gives

$$\left\langle \varphi^{-n}(w)\varphi_1(w_1)...\varphi_{N(w_N)}\right\rangle = \mathcal{L}_{-n}\left\langle \varphi(w)\varphi_1(w_1)...\varphi_{N(w_N)}\right\rangle, \tag{225}$$

where

$$\mathcal{L}_{-n} = \sum_{i=1}^{N}\frac{(n-1)h_i}{(w_i-w)^n} - \frac{1}{(w_i-w)^{n-1}}\partial_{w_i}. \tag{226}$$

In particular, translation invariance gives

$$\mathcal{L}_{-1} = -\sum_{i=1}^{N} \partial_{w_i} = \partial_w. \tag{227}$$

Thus correlators of descendants are fixed by correlators of the corresponding primaries.

5.3.2 OPE and Conformal Blocks

We now consider the OPE of two primaries. Their OPE act on vaccum should give us a state in the Hilbert space grouped into Verma modules. Therefore, we can write the OPE structure universally as a sum:

$$\varphi_1(z,\overline{z})\varphi_2(0,0) = \sum_p \sum_{\ell,\overline{\ell}} C_{12}^{p,\ell,\overline{\ell}} z^{h_p+|\ell|-h_1-h_2} \overline{z}^{\overline{h}_p+|\overline{\ell}|-\overline{h}_1-\overline{h}_2} \varphi_p^{\ell,\overline{\ell}}(0,0). \tag{228}$$

Conformal Ward identities fix the relative contributions of the descendants once the intermediate primary and its OPE coefficient are specified. In particular, the descendant coefficients have the form

$$C_{12}^{p,\ell,\overline{\ell}} = \lambda_{12p} \beta_{12}^{p,\ell} \overline{\beta}_{12}^{p,\overline{\ell}}. \tag{229}$$

The coefficients β depend on the conformal weights and the central charge only, its exact number can be determined by the conformal Ward identities.

One can therefore write the OPE as a sum over primary families:

$$\varphi_1(z,\overline{z})\varphi_2(0,0) = \sum_p \frac{\lambda_{12p}}{z^{h_1+h_2-h_p}\overline{z}^{\overline{h}_1+\overline{h}_2-\overline{h}_p}} \big[\Psi_p(z,\overline{z},0,0)\big]. \tag{230}$$

where the form of $\big[\Psi_p\big]$ is determined by conformal symmetry fully:

$$\big[\Psi_p(z,\overline{z},0,0)\big] = \sum_{\ell,\overline{\ell}} \beta_{12}^{p,\ell} \overline{\beta}_{12}^{p,\overline{\ell}} z^{|\ell|} \overline{z}^{|\overline{\ell}|} \varphi_p^{\ell,\overline{\ell}}(0,0). \tag{231}$$

One may refer to sec. 6.6.3 to understand how we fix all the coefficient β in the OPE. —

Now we consider a four-point function of primary fields.

$$\langle \varphi_1(z_1,\overline{z}_1)\varphi_2(z_2,\overline{z}_2)\varphi_3(z_3,\overline{z}_3)\varphi_4(z_4,\overline{z}_4)\rangle. \tag{232}$$

The four-point function depends on the cross ratios

$$x = \frac{z_{12}z_{34}}{z_{13}z_{24}}, \quad \overline{x} = \frac{\overline{z}_{12}\overline{z}_{34}}{\overline{z}_{13}\overline{z}_{24}}, \tag{233}$$

where $z_{ij} = z_i - z_j$. A global conformal transformation allows us to place the operators at $z_1 = \infty$, $z_2 = 1$, $z_3 = x$, and $z_4 = 0$. Therefore we only need to study a four-point function of the form

$$G_{34}^{21}(x,\overline{x}) = \langle \varphi_1(\infty,\infty)\varphi_2(1,1)\varphi_3(x,\overline{x})\varphi_4(0,0)\rangle. \tag{234}$$

> **Remark**
>
> The order of 1,2 3,4 is important, they notes which operator is at which position.

note that $\varphi_1(\infty,\infty)$ means the limit of $z_1^{2h_1}\overline{z}_1^{2\overline{h}_1}\varphi_1(z_1,\overline{z}_1)$ as $z_1 \to \infty$.

Remark

The explicit relation between the four-point function at any point and G is given as follows:

$$\left\langle \prod_{i=1}^{4} \varphi_{i(z_i,\overline{z}_i)} \right\rangle = P_{h(z_i)} P_{\overline{h}}(\overline{z}_i) G_{34}^{21}(x,\overline{x}), \tag{235}$$

where

$$P_h(z_i) = \left(\frac{z_{24}}{z_{12}z_{14}}\right)^{h_1} \left(\frac{z_{14}}{z_{12}z_{24}}\right)^{h_2} \left(\frac{z_{12}z_{14}}{z_{13}^2 z_{24}}\right)^{h_3} \left(\frac{z_{12}}{z_{14}z_{24}}\right)^{h_4}. \tag{236}$$

The factor $P_{\overline{h}}(\overline{z}_i)$ is obtained by replacing z_{ij} and h_i with $\overline{z}_{ij}$ and $\overline{h}_i$.

In terms of in and out states we can also write:

$$G_{34}^{21}(x,\overline{x}) = \left\langle h_1, \overline{h}_1 \right| \varphi_2(1,1)\varphi_3(x,\overline{x}) \left| h_4, \overline{h}_4 \right\rangle. \tag{237}$$

Applying the OPE to $\varphi_3(x,\overline{x})\varphi_4(0,0)$ we may find that:

$$G_{34}^{21}(x,\overline{x}) = \sum_p \frac{\lambda_{34p}}{x^{h_3+h_4-h_p}\overline{x}^{\overline{h}_3+\overline{h}_4-\overline{h}_p}} \left\langle h_1, \overline{h}_1 \right| \varphi_2(1,1) \left[\Psi_p(x,\overline{x},0,0)\right] |0\rangle. \tag{238}$$

One can introduce a function:

$$\mathcal{A}_{34}^{21}(p|x) = \lambda_{12p}^{-1} x^{h_p-h_3-h_4} \overline{x}^{\overline{h}_p-\overline{h}_3-\overline{h}_4} \left\langle h_1, \overline{h}_1 \right| \varphi_2(1,1) \left[\Psi_p(x,\overline{x},0,0)\right] |0\rangle. \tag{239}$$

and we have:

$$G_{34}^{21}(x,\overline{x}) = \sum_p \lambda_{12p}\lambda_{34p} \mathcal{A}_{34}^{21}(p|x). \tag{240}$$

we recognize that this function is constructed from some 3 point function involving descendants and some holomorphic and antiholomorphic factors. We know that in 2D the holomorphic and antiholomorphic part of 3 point function can be factorized. Therefore the 4 point function itself can be factorized into holomorphic and antiholomorphic part. We can write:

$$G_{34}^{21}(x,\overline{x}) = \sum_p \lambda_{12p}\lambda_{34p} \mathcal{F}_{34}^{21}(p|x)\overline{\mathcal{F}}_{34}^{21}(p|\overline{x}). \quad \text{where} \quad \mathcal{A}_{34}^{21}(p|x) = \mathcal{F}_{34}^{21}(p|x)\overline{\mathcal{F}}_{34}^{21}(p|\overline{x}). \tag{241}$$

We define the holomorphic function $\mathcal{F}_{34}^{21}(p|x)$ as the **holomorphic conformal block**. It is a fixed function constraint by the conformal ward identity and taking out the OPE coefficient of primary fields fixes its normalization. Therefore it is a fully fixed function. The explicit form is given below, The holomorphic conformal block includes the intermediate primary $|h_p\rangle$ and all its Virasoro descendants. If I labels a descendant and $|I|$ is its level, it can be written as

$$\mathcal{F}_{34}^{21}(p|x) = x^{h_p-h_3-h_4} \sum_I \beta_{34}^{p,I} x^{|I|} \frac{\left\langle h_1 \right| \varphi_2(1) L_{-I} \left| h_p \right\rangle}{\left\langle h_1 \right| \varphi_2(1) \left| h_p \right\rangle}. \tag{242}$$

The denominator is just the three-point function of the primary fields by definition.

Remark

Here the overlap means we take only the holomorphic part of the full overlap

It is generally hard to compute the conformal blocks in a close form. However, we can do an expansion and fix the coefficient of each order. Its expansion begins as

$$\mathcal{F}_{34}^{21}(p|x) = x^{h_p-h_3-h_4}(1+F_1x+F_2x^2+...), \quad F_1 = \frac{(h_p+h_2-h_1)(h_p+h_3-h_4)}{2h_p}. \tag{243}$$

Once the external and intermediate conformal weights and the central charge c are fixed, Virasoro symmetry determines the conformal block. The spectrum of intermediate primaries and the three-point coefficients λ are additional data of the CFT.

5.4 Minimal models

Conformal field theories have considerable structure, yet only a few examples follow directly from free-field methods. This motivates the **conformal bootstrap**: combine conformal symmetry, OPE relations, unitarity and other assumptions to constrain or determine a CFT.

A CFT is specified by its spectrum and correlation data, including

$$\big\{(\Delta_i,\ell_i),\lambda_{ijk}\big\}. \tag{244}$$

In two dimensions, the minimal models provide special, highly constrained solutions.

5.4.1 Constraints from unitarity

With $\langle h|h\rangle = 1$, the norm of a level-one descendant is

$$\langle h|L_1L_{-1}|h\rangle = 2h \geq 0, \tag{245}$$

so

$$h \geq 0. \tag{246}$$

To make such calculation easier, we can define the Gram matrix for the descendants. At level k, the descendants are labelled by partitions $I=(n_1,...,n_m)$ with $n_1+...+n_m=k$. We write $|s_I\rangle = L_{-n_1}...L_{-n_m}|h\rangle$. Assuming $\langle h|h\rangle = 1$ and $L_n^\dagger = L_{-n}$, the **Gram matrix** is the matrix of inner products

$$M^k_{IJ}(c,h) = \langle s_I|s_J\rangle. \tag{247}$$

Its size is $p(k)\times p(k)$, where $p(k)$ is the number of partitions of k. The **Kac determinant** is the determinant of this matrix. Up to a nonzero normalization factor A_k, it takes the form

$$\det M^k(c,h) = A_k \prod_{r,s\geq 1, rs\leq k} \big(h-h_{r,s}(c)\big)^{p(k-rs)} \geq 0 \tag{248}$$

Here $h_{r,s}(c)$ are the conformal weights at which a Virasoro module develops a null descendant. Thus, when the Kac determinant vanishes, the Gram matrix is degenerate: some nonzero linear combination of level-k descendants is orthogonal to every state at that level. The factors in the product identify the values of h and c at which this happens.

At level two, consider the Gram matrix for $L_{-1}^2|h\rangle$ and $L_{-2}|h\rangle$:

$$M^2 = \begin{pmatrix} 4h(2h+1) & 6h \\ 6h & 4h+c/2 \end{pmatrix} \geq 0. \tag{249}$$

The trace must be nonnegative:

$$\text{Tr } M^2 = 8h(h+1)+\frac{c}{2} \geq 0. \tag{250}$$

Its determinant factorises as

$$\det M^2 = 32h\big(h-h_{1,1}\big)\big(h-h_{1,2}\big)\big(h-h_{2,1}\big) \geq 0, \tag{251}$$

where

$$h_{1,1} = 0, \qquad h_{1,2}, h_{2,1} = \frac{1}{16}\Big(5 - c \pm \sqrt{(1-c)(25-c)}\Big). \tag{252}$$

The level-two condition excludes a region in the (c, h) plane. At higher levels, the Kac determinant gives further constraints.

> **Remark**
>
> The theories with $c > 1$ are always unitary. To find more interesting results we here consider the case $c < 1$.

Higher levels impose further conditions. The Kac determinant at level k has the form

$$\det M^k = \text{const.} \prod_{r,s \geq 1; rs \leq k} \big(h - h_{r,s}\big)^{P(k-rs)}, \tag{253}$$

where $P(n)$ counts the integer partitions of n, and the roots of the Kac determinant and its relation with the central charge is given by the following form:

$$h_{r,s} = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)}, \qquad c = 1 - \frac{6}{m(m+1)}. \tag{254}$$

In diagram of the (c, h) plane, the region in shaded area is excluded by the Kac determinant.

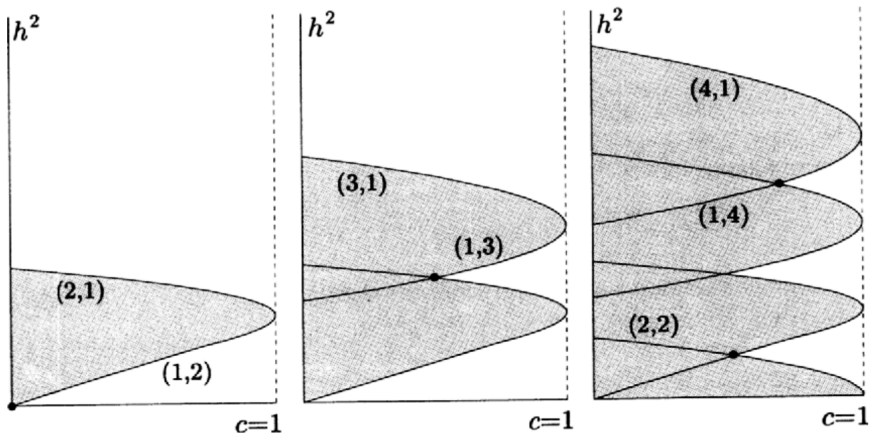


Figure 7.1. The vanishing curves $C_{r,s}$ for levels 2, 3 and 4 (from left to right). The curves at level $l-1$ appear also at level l. The values of r, s are indicated near the curves when they first appear. The black dots are first intersections (defined in the text). The shaded areas correspond to manifestly nonunitary theories.

Figure 1: Kac Determinant

5.4.2 Virasoro Minimal Models

For $c < 1$, one can prove that positivity at every level leaves isolated solutions, such solutions when combined together will give us the minimal models:

$$\mathcal{M}_{m+1,m}, \qquad m = 2, 3, 4, ... \tag{255}$$

The conformal weights and central charge amazingly conincide with the root of the Kac determinant:

$$h_{r,s} = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)}, \qquad c = 1 - \frac{6}{m(m+1)}, \tag{256}$$
$$m = 2, 3, 4, ... \quad r = 1, 2, ..., m-1, \quad s = 1, 2, ..., r.$$

Their allowed primary weights are entries of the Kac table,

$$h = h_{r,s}, \tag{257}$$

notice that there is an identification of the weights in the Kac table, they should be understand as the same primary:

$$h_{r,s} = h_{m-r,m+1-s}. \tag{258}$$

A state $|h\rangle$ in one of these representations has a null descendant at some level k.

5.4.3 Null State and Fusion Rules

For example, the primary $|h_{1,2}\rangle$ has a level-two null state

$$|\chi\rangle = L_{-2} - \frac{3}{2(2h+1)} L_{-1}^2 |h_{1,2}\rangle. \tag{259}$$

The corresponding null field is:

$$\chi(z) = \varphi^{-2}(z) - \frac{3}{2(2h+1)} \partial^2 \varphi(z). \tag{260}$$

One can prove that a correlator containing the corresponding null field vanishes:

$$\langle \chi(z)\varphi_1(z_1)...\varphi_N(z_N)\rangle = 0. \tag{261}$$

Using the differential operators from above, and by the result of correlation function of descendants one can find that:

$$\left(\mathcal{L}_{-2} - \frac{3}{2(2h+1)} \mathcal{L}_{-1}^2\right) \left\langle \varphi(z)\varphi_1(z_1)...\varphi_{N(z_N)}\right\rangle = 0. \tag{262}$$

For a three-point function

$$\left\langle \varphi(z)\varphi_{h_1}(z_1)\varphi_{h_2}(z_2)\right\rangle = \frac{\lambda_{\varphi,h_1,h_2}}{(z-z_1)^{h+h_1-h_2}(z-z_2)^{h+h_2-h_1}(z_1-z_2)^{h_1+h_2-h}}, \tag{263}$$

the null-state equation implies

$$\lambda_{\varphi,h_1,h_2} = 0 \tag{264}$$

unless

$$2(2h+1)(h+2h_2-h_1) = 3(h-h_1+h_2)(h-h_1+h_2+1). \tag{265}$$

These selection rules give out an restriction on what primary fields can appear in the RHS of the OPE of two primary fields. We can denote such relation as a fusion product. If $h_1 = h_{r,s}$, this condition allows

$$h_2 = h_{r,s-1} \quad \text{or} \quad h_2 = h_{r,s+1}. \tag{266}$$

the fusion product or the fusion rule is:

$$\varphi_{1,2} \times \varphi_{r,s} = \varphi_{r,s-1} + \varphi_{r,s+1}, \tag{267}$$

This discussion can be generalized to other chiral primaries. The result is that we can derive the full fusion rules for the minimal models primaries.

5.4.4 Example: the two-dimensional Ising CFT

The Ising model is $\mathcal{M}_{4,3}$. Its primary fields are

$$\sigma : \left(h, \overline{h}\right) = \left(\frac{1}{16}, \frac{1}{16}\right), \quad \varepsilon : \left(h, \overline{h}\right) = \left(\frac{1}{2}, \frac{1}{2}\right), \quad \mathbb{I} : \left(h, \overline{h}\right) = (0, 0). \tag{268}$$

Their fusion rules are

$$\sigma \times \sigma = \mathbb{I} + \varepsilon, \quad \sigma \times \varepsilon = \sigma, \quad \varepsilon \times \varepsilon = \mathbb{I}. \tag{269}$$

The null-state equation also constrains the four-spin correlator. We first focus on the holomorphic part of the correlation function:

$$0 = \left(\sum_{i=2}^{4} \left(\frac{h}{(z_1 - z_i)^2} + \frac{1}{z_1 - z_i} \partial_{z_i} \right) - \frac{3}{2(2h+1)} \partial_{z_1}^2 \right) \langle \sigma(z_1) ... \sigma(z_4) \rangle. \tag{270}$$

Introduce the cross ratio

$$\eta = \frac{z_{12} z_{34}}{z_{13} z_{24}} \tag{271}$$

and write the holomorphic correlator as

$$\langle \sigma(z_1) ... \sigma(z_4) \rangle = \frac{g(\eta)}{z_{12}^{2h} z_{34}^{2h}}. \tag{272}$$

The resulting equation is

$$g''(\eta) + \frac{2 - 8h + 4\eta(h-1)}{3\eta(1-\eta)} g'(\eta) - \frac{2h(2h+1)}{3(1-\eta)^2} g(\eta) = 0, \qquad h = \frac{1}{16}. \tag{273}$$

It has two independent solutions:

$$g_{\pm}(\eta) = \frac{\sqrt{1 \pm \sqrt{\eta}}}{(1-\eta)^{\frac{1}{8}}}. \tag{274}$$

> **Remark**
>
> We can only consider the holomorphic part is due to the fact the constraints constrain the holomorphic and antiholomorphic parts separately.

For the full correlator, it should be a combination of the holomorphic and antiholomorphic parts. A general correlation function of four primary fields can be written as:

$$\langle \sigma(z_1, \overline{z}_1) ... \sigma(z_4, \overline{z}_4) \rangle = \frac{G(\eta, \overline{\eta})}{|z_{12}|^{4h} |z_{34}|^{4h}}. \tag{275}$$

The most general combination of the two holomorphic and two antiholomorphic solutions is

$$G(\eta, \overline{\eta}) = C_{++} g_+(\eta) g_+(\overline{\eta}) + C_{+-} g_+(\eta) g_-(\overline{\eta}) + C_{-+} g_-(\eta) g_+(\overline{\eta}) + C_{--} g_-(\eta) g_-(\overline{\eta}). \tag{276}$$

The full correlator must also obey:

- Single-valuedness under monodromy around $\eta = 0$ and $\eta = 1$.
- Crossing symmetry.
- Consistency with the OPE.

In this example, single-valuedness fixes the combination, up to an overall real constant A:

$$G(\eta,\overline{\eta}) = A\frac{|1+\sqrt{\eta}|+|1-\sqrt{\eta}|}{2|1+\eta|^{\frac{1}{4}}}. \tag{277}$$

> **Remark**
>
> This constant can not be fixed by all above constraints. Cause such constraints are differential equations and the solution is always up to a constant.

We know that the correlation function at $0,1,x,\infty$ can be expand in terms of the conformal blocks:

$$G(\eta,\overline{\eta})\ |z_{12}|^{-4h}\ |z_{34}|^{-4h} = \sum_p \lambda^2_{\sigma\sigma p}\mathcal{F}_p(\eta)\overline{\mathcal{F}}_p(\overline{\eta}). \tag{278}$$

with a normalization of $A=1$ we can read off the OPE coefficients from the expansion of the conformal blocks:

$$\lambda_{\sigma\sigma\mathbb{I}} = 1, \quad \lambda_{\sigma\sigma\varepsilon} = \frac{1}{2}. \tag{279}$$

> **Remark**
>
> The OPE coefficients are A dependent. One may understand this as we have a freedom to rescale the primary fields. The conformal blocks are fixed functions, they are independent of the normalization of the primary fields.

Bibliography